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Digital financial platform engagement and financial inclusion in the Philippines: Insights on AI deployment and policy implications

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Digital Financial Platform Engagement and Financial Inclusion in the Philippines: Insights on AI Deployment and Policy Implications

*Nikka C. Pesa, Mary Grace R. Agner, and
Rutcher M. Lacaza*

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Digital Financial Platform Engagement and Financial Inclusion
in the Philippines: Insights on AI Deployment and Policy
Implications

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Abstract

This paper examines how digital financial platform engagement influences financial inclusion in the Philippines and explores how financial institutions are deploying artificial intelligence (AI) in enhancing these platforms. It combines nationally representative data from the World Bank Global Findex (2021), institutional indicators from the International Monetary Fund (IMF) Financial Access Survey (2016–2024), and qualitative insights from financial institutions and policy experts. From the demand-side a Digital Financial Engagement Index was constructed based on individuals' usage of mobile payments, online banking, and digital financial services. Results reveal that digital financial engagement is a strong and consistent determinant of financial inclusion: individuals who actively use digital financial platforms are significantly more likely to own and use formal financial accounts. However, several barriers persist. The lack of money, high perceived costs, documentation issues, and low trust remain key constraints among Filipino adults, particularly among low-income groups. From the supply side, institutional data from the IMF Financial Access Survey (2016–2024) and insights from Key Informant Interviews (KIIs) reveal that financial infrastructure has expanded moderately. While AI adoption remains nascent and largely concentrated among large, digitally advanced financial institutions, AI technologies such as fraud detection, credit scoring, and chatbots are increasingly embedded in the digital platforms where consumer transact. In contrast, smaller cooperatives and savings and loan associations continue to face structural and resource-related barriers to adoption. The convergence of demand- and supply-side evidence highlights that digital financial engagement serves both as a driver and an indicator of financial inclusion, while AI adoption within these platforms has potential to further enhance access, security, and user experience. Overall, this study provides new empirical evidence that digital financial engagement significantly enhances financial inclusion in the Philippines. The study also documents how financial institutions are strategically deploying AI within these digital platforms, providing insights for responsible AI adoption that can support inclusive financial services. By strengthening digital infrastructure, promoting financial and digital literacy, and ensuring responsible AI adoption, the Philippines can transform digital financial platforms into a true catalyst for inclusive and sustainable growth.

Keywords: digital financial engagement, artificial intelligence, financial inclusion, digital platforms, Philippines

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Digital Financial Platform Engagement and Financial Inclusion in the Philippines: Insights on AI Deployment and Policy Implications

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1. Introduction

1.1 Background of the Study

Access to formal financial accounts in the Philippines has developed over the years. With the rise of the internet and the emergence of electronic banking in the 1980's, Filipinos have acquired formal financial accounts through automated teller machines (ATMs), internet banking, and digital wallets (Bangko Sentral ng Pilipinas n.d.). A basic measure of financial inclusion is having possession of an account within a recognized institution, which encompasses banks, e-money issuers, and microfinance institutions (Bangko Sentral ng Pilipinas 2021). Years of economic growth have paved the way for several underprivileged Filipinos to become part of the middle class which led them to experience the benefits of its increased purchasing power (De Gantès et al. 2023). Also, they note that a young techy-savvy consumer base is fuelling a heightened demand for innovative financial accounts and services. According to the recent 2021 Financial Inclusion Survey, formal account ownership among Filipinos has drastically improved, reaching 56% in 2021 compared to 29% in 2019 with e-money accounts now being the most owned. However, when comparing it to selected economies in the Association of Southeast Asian region such as Singapore (97.55%), Thailand (95.58%), Malaysia (88.37%), and Viet Nam (56.27%), the World Bank (2021) reports that the Philippines has relatively higher percentage of unbanked and unserved population (51.37%). Leveraging digital technology can narrow the gap of this segment in accessing and availing financial services, purchasing products and services, as well as engaging in other transactions.

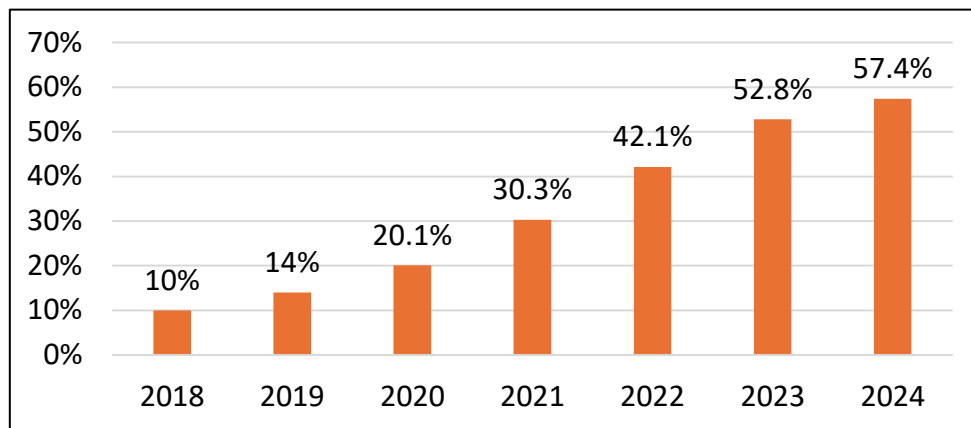
As the Philippine financial system continues to adapt with emerging technology, traditional universal and commercial banks are now going heavy in offering online/digital banking for their consumers following BSP's Digital Payments Transformation Roadmap 2020-2023. Under this roadmap, one of the BSP's targets is to have 50% of the total retail transaction volume be digital. As seen in Figure 1, the BSP reported that 57.4% of the total volume of retail payments (5,756.5 billion) in 2024 came from digital payments (3,307 billion), with non-digital payments now holding lesser of the share by 2,449.5 billion (Bangko Sentral ng Pilipinas n.d.). Targets for digital payments have been increased to 52 to 54% under the Philippine Development Plan of 2023 to 2028. Figure 1 also shows that these targets have been met since digital retail payments accounted for 57.4% of total transaction volume according to the 2024 e-payments measurement conducted by the BSP. Digital payments accounted for 3.307 billion while non-digital payments contributed 2.449 billion to the total transactions volume.

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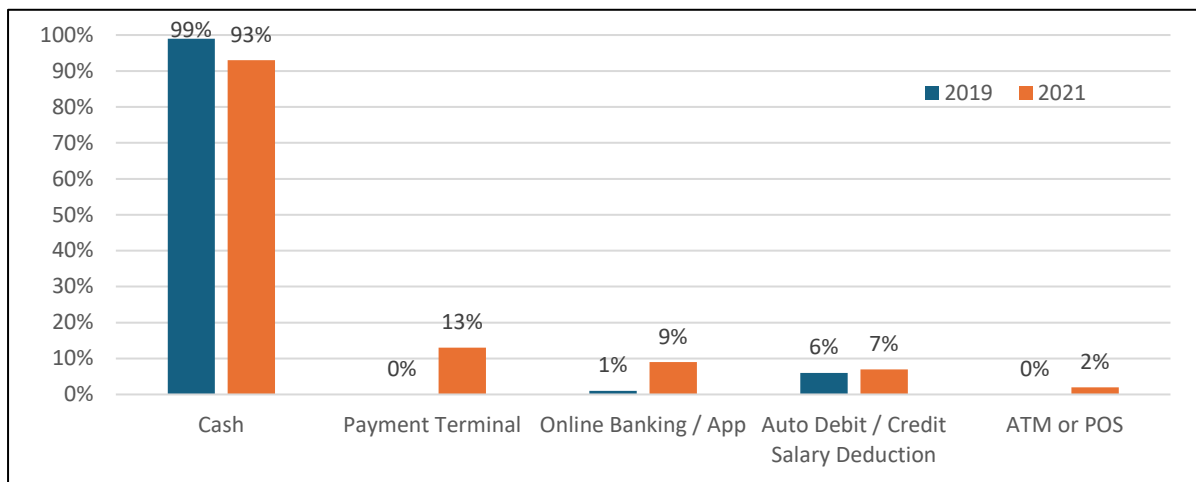
Figure 1. Share of Digital Payments to Total Retail Payments, 2018 - 2024



Source: Bangko Sentral ng Pilipinas 2024

Moreover, the 2021 Financial Inclusion Survey reported cash as the leading mode of payment, as seen in Figure 2. While this is the case, with the increasing number of Filipinos with bank accounts and emergence of digital banks, digital retail payments also increased to 42% in 2022 (Bangko Sentral ng Pilipinas, n.d.). The BSP targeted 50% for 2023 given its recent issuance of licenses to 258 digital payment providers. Because of the increase in digital transactions, banking fraud and cybercrime has also increased (De Vera 2022). This is the reason why Filipinos are hesitant to put money in banks (Lim 2024), reluctant to use credit products (Jocson 2023; Yu 2023), and not privy to engaging in more digital transactions (De Gantès et al. 2023).

Figure 2. Mode of Payment used by Adult Filipino Consumers, 2019 and 2021



Note: Respondents can report more than one response. Base population – Among those who made payments (84% of Filipino adults)

Source: Bangko Sentral ng Pilipinas 2021

Artificial intelligence (AI) and machine learning (ML) are gaining prominence across sectors, especially in financial services (OECD 2021). Khsetri (2021) highlighted that the fast advancements in AI and ML are instigating political, economic, and social transformations in developing economies. According to (Khsetri 2021), AI's revolutionary effects have reshaped every sector of the economy, including financial services, and it has led to a paradigm shift in enhancing financial inclusion within an economy. Moreover, adoption of AI and ML in financial sector offers financial deepening, efficiency, transformation of financial services and consumers' interaction with the financial services ecosystem (Khsetri 2021). On the other hand,

one should take into account as well that this could likely lead to undesirable effects, such as, data privacy vulnerabilities, and unfair outcomes for certain consumers like women (Rizzi and Alvarez 2023), and intensifying the digital divide between advanced and developing economies (Boukherouaa et al 2021). With these considerations in mind, and building on the discussion of De Gantès et al. (2023) and the OECD (2021), it is worthwhile to explore the AI deployment in the Philippine financial sector, including its impact both on those who are already financially included and on those Filipinos who have yet to be reached by the banking sector and whose confidence in the financial sector must be increased.

While the potential of AI to advance financial inclusion is clear, studying its direct impact faces significant methodological challenges. Large-scale household surveys do not yet capture direct measures of AI awareness, usage, or literacy (World Bank 2021). Moreover, since AI technologies are often embedded invisibly within digital financial platforms, consumers may use AI-powered fraud detection or credit scoring without realizing it. To address these constraints, this study adopts a two-sided approach. On the demand side, the analysis examines how engagement with digital financial platforms – the primary channels through which AI-enabled services are delivered – influences financial inclusion. On the supply side, it investigates how financial institutions deploy AI within these platforms. Collectively, this framework captures both the behavioral foundations of AI-enabled financial inclusion, reflected in users’ digital engagement, and the institutional realities of AI adoption.

1.2 Rationale of the Study

The Philippines has made substantial progress in advancing financial inclusion, with formal account ownership rising significantly in recent years. Despite these gains, a considerable segment of the population remains either unbanked or underbanked due to barriers such as cost, distance, lack of documentation, and low trust in financial institutions. At the same time, the proliferation of digital payments and e-money accounts highlights both the potential and the complexity of ensuring inclusive and secure participation in the financial system. Artificial intelligence (AI), increasingly embedded within digital financial platforms, offers new tools to address these challenges by improving credit assessment, fraud detection, and personalized financial services. However, its adoption also raises concerns about data privacy, consumer protection, and the risk of deepening inequalities if safeguards are not in place. This study is therefore undertaken to examine how digital financial platform engagement influences financial inclusion and how artificial intelligence can be strategically deployed within these platforms to enhance accessibility, safety and security, drawing insights from both consumers’ behavior and institutional readiness.

1.3 Statement of the Problem

Recent developments in the Philippine financial sector have indicated that the economy is undergoing a digital banking revolution (De Gantès et al. 2023) where technology is seen as a vehicle to boost the financial literacy of Filipinos (Lim 2024). Specifically, the emergence of AI, including ML, and big data, in the financial sector have opened opportunities for financial institutions to enhance their presence and reach to Filipino households across all demographic classifications, posed challenges on the safety and security of banking and making financial transactions online, and created implications for policymakers to craft frameworks towards regulation, safety, and security in the use of technology to make transactions. Against this backdrop, this study poses the following research question: ***how does engagement with digital financial platforms influence financial inclusion in the Philippines, and how are financial***

institutions deploying artificial intelligence within these platforms aiming to enhance accessibility, safety and security?

1.4 General and Specific Objectives

In addressing the above research question, the study's general objective is to examine how digital financial platform engagement influences financial inclusion among Filipino households, and to inform policymakers how AI can be strategically deployed within these platforms to enhance access, trust, safety, and security in digital financial transactions. The following specific objectives are also set.

1. to examine how digital financial platform engagement affects financial inclusion outcomes from the demand side, and to assess the current state and potential of AI adoption within financial institutions from the supply side;
2. to provide empirical evidence on the relationship between digital financial engagement and financial inclusion, and on the opportunities and challenges of AI adoption in the Philippine financial sector;
3. identify barriers to financial inclusion, including safety, security, trust, and cost concerns and assess how AI deployment in digital financial platforms can help address these barriers; and
4. to inform policymakers on strategies for strengthening digital financial infrastructure, promoting digital and financial literacy, and ensuring responsible AI deployment to advance inclusive and secure financial services in the Philippines.

1.5 Significance of the Study

Employing advanced digital technology in the financial sector has much opportunity for advancing the Philippines's growth (Open Gov Asia 2021). For instance, the Bangko Sentral ng Pilipinas (2018) highlighted the advantages of digital financial inclusion provides greater access to formal financial services, economic empowerment and reduced risks and costs associated with cash. The said institution emphasized its ability to access untapped markets, facilitate asset accumulation, boost economic participation, decrease cash handling costs, and mitigate the elevated risks of theft and fraudulent activities linked to the informal economy. The Philippines is experiencing one of the fastest rates of population growth globally, and the bankable population is expected to surge from 65 million in 2022 to 85 million by 2030, marking a 30% growth (De Gantès et al. 2023).

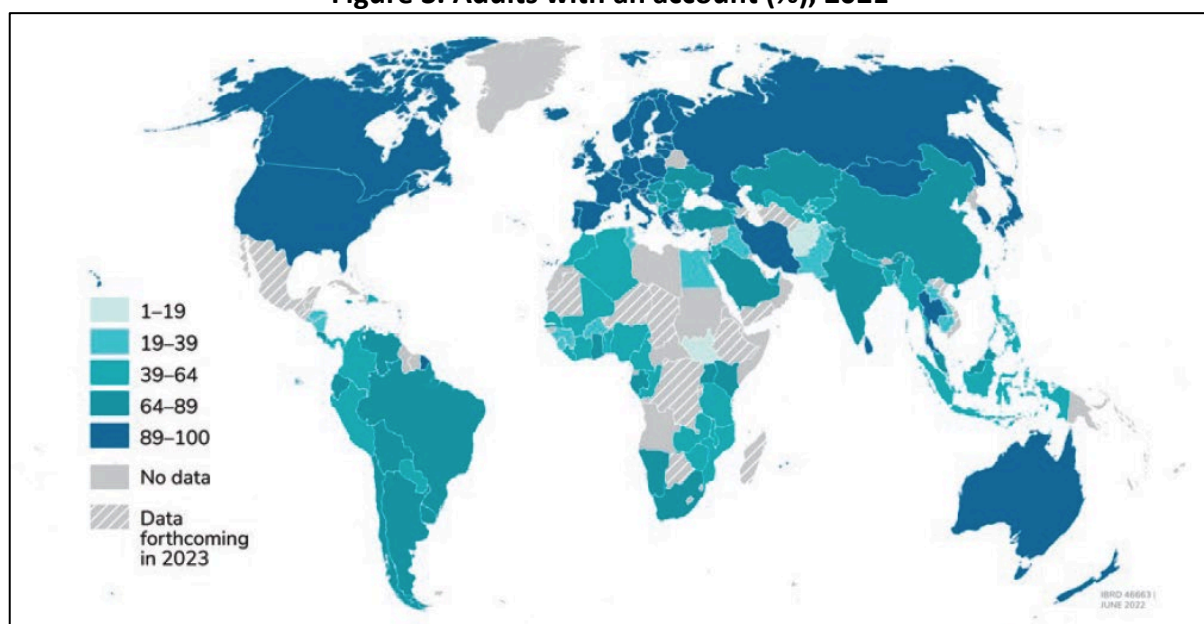
Understanding the relationship between digital financial platform engagement and financial inclusion, and how artificial intelligence can enhance these platforms, is therefore critical for maximizing these opportunities. This paper makes several contributions to the growing literature on digital finance, AI, and financial inclusion in the Philippines. First, methodologically, the study demonstrates how to research the role of emerging technologies when direct survey measures are unavailable. By examining digital financial platform engagement from nationally representative survey data and triangulating this with institutional evidence of AI deployment through key informant interviews and administrative data, the study provides a comprehensive picture of how technology is reshaping financial inclusion. This two-sided approach addresses the challenge of studying AI's effects before survey instruments fully capture the phenomenon.

Second, empirically, the study provides new evidence on the determinants of financial inclusion in the Philippines, with particular attention to the role of digital financial engagement and the barriers that persist despite technological advancement. The findings directly inform the BSP’s efforts to achieve two of its four key outcomes: reducing disparities in financial inclusion and empowering more consumers to be financially capable (BSP, n.d.).

Third, from a policy perspective, the study offers actionable insights on how AI can be strategically deployed within digital financial platforms to enhance accessibility, safety, and security. As pointed out by the OECD (2021), policymakers play a crucial role in fostering innovation within the financial sector, simultaneously safeguarding the interests of financial consumers and investors, and maintaining fairness, orderliness, and transparency in the markets associated with such products and services. This study provides evidence-based guidance for policymakers on strengthening digital infrastructure, promoting digital and financial literacy, and ensuring responsible AI adoption, all of which are essential for translating digital financial engagement into truly inclusive outcomes.

Aside from its relevance to the Philippines, this study, especially its recommended strategies to policy makers, may also be helpful to other countries, especially those with significant unbanked and unserved populations. Around 76% of the world’s population have an account either at a bank or a financial institution, including mobile money service providers. This percentage may suggest good progress to gauge financial inclusion globally; however, considering the regional or economy-level view, there exists a wide variation worldwide. Among the 123 surveyed economies of the Global Financial Inclusion (Global Findex) Database, “account ownership ranges from just 6% in South Sudan to universal ownership in high-income economies such as Canada, Germany, and the United Kingdom” (Demirgüç-Kunt et al. 2021, p. 15). Figure 3 below highlights further this discrepancy. Different economies may have varied ways of addressing this issue on access to financial services; however, this study on the Philippines can present useful insights to other countries, especially those facing similar challenges in leveraging digital platforms and AI to advance financial inclusion.

Figure 3. Adults with an account (%), 2021



Source: Demirgüç-Kunt et al. 2022

1.6 Scope and Limitation

Given the nature and objectives of this study, the authors examine how digital financial platform engagement influences financial inclusion, and how artificial intelligence is being deployed within these platforms by financial institutions. The study adopts an Explanatory Sequential Mixed Method Research approach that addresses several important methodological limitations.

On the demand side the study uses data from the World Bank's Global Financial Inclusion (Global Findex) Database is utilized. This database comprises nationally representative surveys conducted among approximately 128,000 adults across 123 economies. The study specifically uses the 2021 Global Findex Database for the Philippines. However, a key limitation is that the Global Findex, like most large-scale household surveys, does not include direct questions about AI awareness, knowledge, literacy, or usage (World Bank 2021). Therefore, the study constructs a Digital Financial Engagement Index based on individuals' usage of mobile payments, online banking, and digital financial services. This index measures engagement with the digital platforms through which AI-enabled services are increasingly delivered, rather than direct AI exposure or awareness. This limitation means that the demand-side analysis captures the behavioral foundation for AI-enabled financial inclusion (digital platform usage) rather than AI interaction itself. Moreover, actual AI exposure varies across platforms and institutions. An individual using GCash may encounter different AI features than someone using a small rural bank's digital service, even though both would score similarly on the Digital Financial Engagement Index. The index cannot distinguish between high-AI and low-AI platforms based on available survey data. Future research with purpose-designed surveys can address this limitation by directly measuring AI awareness and platform-specific usage patterns.

To complement the demand-side analysis and establish the connection to AI, the supply-side perspective is captured using the IMF's Financial Access Survey (2016–2024), alongside key informant interviews with policy experts and private financial institutions were conducted to capture the supply-side perspective. Findings from both approaches are triangulated to enrich and contextualize the quantitative results. This triangulation allows the study to demonstrate how digital financial engagement (measured on the demand side) relates to financial inclusion outcomes, while documenting the AI deployment patterns (observed on the supply side) within the digital platforms that users engage with. This two-sided approach addresses the methodological challenge of studying emerging technologies before comprehensive survey instruments exist, though it acknowledges the limitations of indirect measurement. The study does not examine the technical specifications or algorithmic details of specific AI systems (e.g., neural network architectures, training datasets, or model performance metrics), as the emphasis is on behavioral and institutional perspectives rather than computer science or engineering dimensions of AI deployment.

The study also covers institutional and regulatory aspects related to AI adoption, particularly issues concerning regulatory capacity, stakeholder engagement, legal frameworks, and consumer education. These areas are examined to understand how they influence the integrity and security of financial systems in the context of AI-driven financial inclusion. With these scope parameters and limitations in mind, the paper aims to guide Philippine policymakers in crafting evidence-based policies and institutional frameworks that strengthen digital financial infrastructure, promote digital and financial literacy, and ensure responsible AI deployment to expand financial access and inclusion.

2. Review of Related Literature and Research Gap

This literature review examines existing research on digital financial platforms and financial inclusion. The review begins with theoretical frameworks for technology adoption (Section 2.1), followed by empirical evidence on how digital financial platforms and AI influence financial inclusion from both consumer and institutional perspectives (Section 2.2). The review then examines digital infrastructure and AI readiness in the Philippine context (Section 2.3), specific applications of artificial intelligence in financial services (Section 2.4); variables commonly used in financial inclusion studies (Section 2.5), and existing policy frameworks (Section 2.6). The presentation of the research gap follows.

2.1 Theoretical Frameworks for Technology Adoption

The Technology Acceptance Model (TAM) proposed by Davis (1989) has been instrumental in explaining users' behavioral intentions toward adopting new technologies. It posits individuals who perceive a technology as useful and easy to use are more likely to develop a positive attitude toward it. In other words, the perceived usefulness and perceived ease of use are the key determinants influencing attitudes and intentions to adopt a technology. In the context of digital finance, TAM has been widely used to analyze customers' intention to adopt mobile banking, e-wallets, and other financial technologies (Rahman et al. 2023).

However, while TAM provides valuable insights, it tends to underrepresent the roles of trust, perceived risk, and data security, factors that are increasingly critical in the adoption of technologies associated with financial data and user privacy (Manser Payne et al. 2018). These limitations are particularly relevant when studying artificial intelligence (AI) applications in finance, where concerns about data privacy, algorithmic transparency, and cybersecurity strongly shape adoption behavior.

To address these gaps, combining TAM with Diffusion of Innovations (DOI) theory has been widely adopted by researchers (Rogers 1962). Within this context, relative advantage is a key factor defined as the degree to which a new technology is perceived by users as providing greater benefits than existing alternatives. By highlighting the added value introduced by the new technology, relative advantage enhances TAM's perceived usefulness. In the context of AI-enabled financial services, this relative advantage may be reflected in improved transaction security, personalized financial advice, and enhanced fraud detection, all of which strengthen user confidence and increase the intention to adopt.

2.2 Digital Infrastructure and Institutional Readiness

The Philippines' capacity to expand digital financial services and adopt advanced technologies depends fundamentally on its digital infrastructure and institutional readiness. Multiple indices and assessments provide insights into the country's current position and challenges in this regard. The International Monetary Fund in 2023 constructed an AI Preparedness Index (APII) based on macro-structural indicators. The indicators cover 174 countries' digital infrastructure; human capital and labor market policies; innovation and economic integration; and regulation and ethics. As shown in Table 1, the Philippines recorded an overall score of 0.50 which was slightly higher than Viet Nam (0.48) but relatively lower than Indonesia (0.52), Thailand (0.54), and Singapore (0.80). Moreover, the Philippines recorded the following values: Digital Infrastructure (0.10), Innovation and Economic Integration (0.12), Human Capital and Labor Market Policies (0.15), and Regulation and Ethics (0.12).

Broader ASEAN-wide indicators provide additional context on the country's digital and AI landscape. As presented in Tables 1 and 2, the Philippines occupies a middle position across multiple dimensions, not only in AI preparedness but also in digital readiness, financial inclusion indicators, and digital economy size. What distinguishes the country, however, is the contrast between its very high consumer interest in digital technologies and comparatively weak institutional capacity. While Filipinos rank among the highest in the region in digital technology engagement search activity, the ASEAN Digital Integration Index shows that digital infrastructure and innovation capability remain the country's weakest pillars, which are precisely the areas needed for financial institutions to design, deploy, and scale technology-enabled services. This structural "curiosity-capacity gap" helps explain institutional adoption of advanced technologies such as AI in the financial sector remains gradual despite strong consumer-driven digital engagement.

Table 1. ASEAN AI & Digital Readiness

Country	AIPI	Oxford AI	GMV 2024 (\$B)	Growth %	Account %	Digital Pay %	AI Search Interest	Mobile Money %
Singapore	0.80	82.0	26B	15%	98%	86%	Very High	31%
Malaysia	0.58	68.7	23B	18%	88%	72%	Very High	17%
Thailand	0.54	63.0	53B	23%	96%	69%	High	11%
Indonesia	0.52	61.0	82B	25%	52%	39%	Moderate	10%
Philippines	0.50	52.0	28B	20%	56%	55%	Very High	30%
Vietnam	0.48	54.5	40B	22%	56%	47%	Moderate	19%
Cambodia	—	31.9	4B	18%	—	—	Low	—
Laos	—	33.1	2B	16%	—	—	Low	—
Myanmar	—	30.9	3B	12%	—	—	Very Low	—
Note. AIPI = Artificial Intelligence Preparedness Index from the International Monetary Fund (0–1 scale). Oxford AI refers to the Oxford Government AI Readiness Index (0–100 scale). Digital GMV represents the estimated 2024 gross merchandise value across e-commerce, transport, travel, food delivery, and online media sectors (Google–Temasek–Bain, 2024). Account ownership and digital payment usage percentages are based on the World Bank Global Findex 2021. AI Search Interest reflects relative Google search volume for AI-related queries. Mobile Money % refers to adults who used mobile money services in the past year. Dashes (—) indicate data not available. Sources. IMF (2023); Oxford Insights (2023); Google–Temasek–Bain e-Conomy SEA Report (2024); World Bank Global Findex (2021); ASEAN Digital Integration Index (2021–2023).								

Table 2. ASEAN Digital Integration Index

Country	Overall	Infrastructure	Cybersecurity	Digital Payments	Skills / Innovation	E-Commerce
Singapore	High	Excellent	Excellent	Excellent	Excellent	Excellent
Malaysia	High	Good	Good	Good	Good	Good
Thailand	Mod-Good	Good	Moderate	Good	Moderate	Good
Indonesia	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
Philippines	Moderate	Weak	Moderate	Moderate	Weak	Moderate
Vietnam	Moderate	Moderate	Moderate	Moderate	Weak	Moderate
Cambodia	Low	Weak	Weak	Weak	Weak	Weak
Laos	Low	Weak	Weak	Weak	Weak	Weak
Myanmar	Very Low	Very Weak	Very Weak	Weak	Very Weak	Weak
Note. Overall Score reflects aggregate performance across digital infrastructure, cybersecurity and data protection, digital payments, digital skills and innovation, and e-commerce readiness. Ratings follow ASEAN classification categories (Excellent, Good, Moderate, Weak, Very Weak). Digital Infrastructure evaluates affordability, reliability, and coverage of connectivity. Cybersecurity & Data Protection measures regulatory frameworks and digital trust systems. Digital Payments assesses usage and interoperability. Skills/Innovation captures workforce digital readiness and innovation capacity. Source. ASEAN Digital Integration Index, ASEAN Secretariat (2021–2023).						

On the other hand, the Bangko Sentral ng Pilipinas (2024) reported that the adoption of AI within the Philippine financial sector is currently limited to specific, narrow applications. The BSP noted that financial institutions in the country have reached different stages of maturity in terms of AI adoption and governance. Based on their survey of financial institutions, the most common AI application is identifying Patterns and Anomalies, which primarily used in Fraud and Anti-Money Laundering (AML) systems. The report also indicated that most of these institutions remain conservative in their willingness to take risks associated with AI. There are existing Philippine studies examining the adoption of AI in terms of customer satisfaction with mobile banking services (Carandang 2023 and Lim and Parrocho 2024), banking operations and performance (Jaca et al. 2025), and the perceptions of customers and employees of banking service and experience (Bragas et al. 2025). Despite these, existing empirical research on how digital financial platform engagement influences financial inclusion using nationally representative data, and how financial institutions are deploying AI in the Philippines remains limited. Table 3 shows the summary of the international and local studies on AI in the financial sector.

Table 3. Literature Matrix

Author(s)	Country	Data Set	Methodology	General Findings
Al-Okaily et al. (2024)	Jordan	Field Survey (270 employees)	Partial Least Squares-Structural Equation Modelling	Level of awareness, security, and privacy can influence finance through mobile penetrations among people
Bangko Sentral ng Pilipinas (2024)	Philippines	Survey (48 institutions)	Qualitative (Consultations)	Risks specific to AI are yet to be addressed. AI adoption is confined to narrow usages. Philippine institutions are at different maturity levels in AI adoption
Balaratnam et al. (2025)	Several	Survey (8,400 respondents)	Mixed Method	AI has the highest uptake in digital payments; less willingness use AI when the financial stakes are higher
Bragas et al. (2025)	Philippines (Metro Manila and Urban Areas)	Survey (161 respondents)	Mixed Methods Research	AI is viewed by employees as helpful facilitator of efforts to automate everyday processes, reduce human error, and enable deeper engagement with customers. Even though there are customers who are deemed loyal and satisfied with the use of AI, various concerns persist among the bank customers on technological innovation to protect their data.
Carandang (2023)	Philippines (Lipa City)	Survey (150 respondents)	Descriptive Analysis and Regression	Technology-based determinants (attitudes toward artificial intelligence, relative advantage, security in specific mobile banking activities, and perceived trust in mobile banking) have significant effect on mobile banking usage
Fazal et al. (2023)	Several	Systematic Literature Review	Systematic Literature Review	AI is important for achieving financial inclusion, but there needs to be more AI and ML methodologies in research.
How et al. (2020)	United States	Profiles of prospective customers from 45,211 contact encounters	AI-Based BN Approach	No-coding human-centric approach can be used in predicting intentions toward financial products
Jaca et al. (2025)	Philippines (Taguig City)	Survey (100 respondents)	Descriptive-correlational Research	AI integration has positively impacted banking performance, particularly in operational efficiency, risk management, and strategic decision-making.

				Customer satisfaction remains an area needing improvement
Jackson and Cepa (2025)	United States	Survey (7,255 respondents)	Survey collection	Lack of awareness is a primary obstacle, suggesting that improving familiarity with AI tools is key to increasing their adoption
Kaur and Dharmadhikari (2023)	India	Bank Nifty Stocks; Secondary Data	Secondary Data Evidence	The major use of AI is in form of chat bots, loan appraisals, risk management, credit assessments, bank enquiries, product applications, payment transfers, ATM-related services, and personalized offers
Lim and Parrocho (2024)	Philippines	Survey (486 respondents)	Confirmatory Factor Analysis, Structural Equation Modelling, Mediation Analysis	Perceived intelligence enhanced the propensity of users to adopt mobile banking apps through impact task-technology fit, perceived cost, and trust. Anthropomorphism exhibited a positive effect on users' perceived cost.
Lim et al. (2025)	Malaysia	Survey (192 respondents)	Technology Acceptance Model	Since using banking apps requires users to be tech-savvy, awareness of technology is crucial prerequisite for adoption
Manrai et al. (2021)	India	Survey (568 respondents)	Structural Equation Modelling	Effort expectancy, habit, facilitating conditions, perceived competence are significant determinants of adoption of digital payments
Mhlanga (2020)	Several	Peer-reviewed journals, reports and other authoritative documents	Conceptual and documentary analysis	AI has a strong influence on digital financial inclusion in areas related to risk detection, measurement and management, addressing the problem of information asymmetry, availing customer support and helpdesk through chatbots and fraud detection and cybersecurity.
Noreen et al. (2023)	Several Asian countries	Survey (799 respondents)	Multiple Regression and ANOVA Analysis	Awareness, attitude, subjective norms, perceived usefulness, and knowledge of artificial intelligence had a significant and positive relationship with intention to adopt AI in the banking sector
Rahim et al. (2023)	Malaysia	Survey (418 respondents)	Structural Equation Modelling	Fintech innovations adoption is possible through performance expectancy, effort expectancy, social influence and facilitating conditions.

Telukdarie and Mungar (2023)	Several	Literature Analysis and Systems Dynamic framework	Literature Analysis and Systems Dynamic framework	Financial inclusion is identified as key driver in helping these economies reap the benefits of their growth. Artificial Intelligence may serve as a power tool to obtain access to financial institutions.
Óskarsdóttir et al. (2019)	Belgium	Call-detail records, credit and debit account information of customers	Social networks and influence propagation	Machine learning, a subset of Artificial Intelligence, can be used for evaluating credit scoring and enhancing financial inclusion. Combining call-detail records with traditional data in credit scoring models significantly increases their performance when measured in AUC.

Source: Constructed by the authors

2.3 Digital Financial Platforms and Financial Inclusion

Digital financial platforms, including mobile banking, e-wallets, online payment systems, and digital credit services, have emerged as critical channels for expanding financial inclusion. These platforms reduce transaction costs, improve accessibility, and enable financial services to reach populations previously excluded from the formal financial system. In a systematic literature review, Fazal, Ahmed, and Nisar (2023) examined how technological innovations, including artificial intelligence techniques are enhancing financial inclusion in developing economies. Their study indicated that qualitative studies center on the contributing knowledge regarding AI risks and challenges, while quantitative studies focused on people's behaviors towards AI adoption. For instance, the authors identified a study that recommended facilitating conditions to induce change in semi-rural women toward technology adoption (Manrai et al. 2021). Additionally, they highlighted that level of awareness, security, and privacy can influence finance through mobile penetrations among people (Al-Okaily et al. 2024). Lastly, they revealed that fintech innovations adoption is possible through performance expectancy, effort expectancy, social influence and facilitating conditions (Rahim et al. 2023).

From the consumer perspective, recent research has examined factors influencing adoption of digital financial services and AI-enabled tools within these platforms. Jackson and Cepa (2025) found that while consumers recognize the possibilities AI presents, many remain concerned about AI-generated misinformation and data security. Based on the Financial Health Pulse Survey, most consumers (including Asian, Black, Latinx, and White groups) rely on traditional sources like banks, personal finance websites, and finance professionals for financial information, with AI tools such as general chatbots and robo-advisors remaining less commonly used (Jackson and Cepa 2025). The authors concluded that a lack of awareness is a primary obstacle, suggesting that improving familiarity with AI tools is key to increasing their adoption.

Several studies have confirmed the importance of AI and technology knowledge in driving adoption. Noreen, Shafique, Ahmed, and Ashfarq (2023) found that consumers' knowledge of artificial intelligence technology had a significant and positive relationship with the intention of AI adoption in the banking sector. Moreover, a study by Lim, Low, Yeo, and Tan (2025) found that Generation Z individuals are willing to adopt AI in banking, indicating their

preference for using AI-enabled technology for managing financial services over consulting a human financial advisor. The authors further suggest that since using banking apps requires users to be tech-savvy, awareness of technology is crucial prerequisite for adoption. Lastly, Balaratnam, Palaniappan, Merchant, Da Gama-Rose, Pandita, Cherukupalli, and Kanwar (2025) constructed an AI Inclination Index to measure the consumer preferences of AI adoption in banking and finance. One measurement being captured by the AI Inclination Index is Technology Comfort which gauges “consumers’ attitude toward trying the latest technology, as well as their comfort with using their smartphone in the purchase journey”. The research revealed that AI has the highest uptake in digital payments, where consumers are accustomed to its background role in improving security and convenience. Yet, consumers show less willingness to complete a transaction using AI when the financial stakes are higher, with Technology Comfort scores lower during the "Buy" phase compared to the "Learn" and "Use" phases of engagement.

From the institutional perspective, financial service providers have increasingly deployed AI and related technologies within digital platforms to improve service delivery, manage risks, and expand reach to underserved populations. Kaur and Dharmadhikari (2023) examined how major banks promote financial inclusion through the implementation of artificial intelligence (AI)-based products and services. Their study indicated that AI is predominantly utilized in the form of chatbots, loan evaluations, risk management, credit assessments, bank enquiries, product applications, payment transfers, ATM-related services, and personalized offers (Kaur and Dharmadhikari 2023). Kandpal and Khalaf (2020) stressed that AI usage in banking results in cost-effective and efficient solutions that cater financial services to the financially excluded population. They also contended that models driven by big data could be employed for psychometric assessments, capturing data that aids in predicting loan repayment behavior, applicants' beliefs, performance, attitudes, and integrity. Mhlanga (2020) revealed that AI has a strong influence on digital financial inclusion associated with risk detection, measurement and management, addressing the problem of information asymmetry, availing customer support and helpdesk through chatbots, and fraud detection and cybersecurity.

2.4 AI Applications in Financial Services

Enhancing financial inclusion through digital tools and technologies and uplifting the welfare of households and communities have been employed by most countries (Ozili 2021). Kaur and Dharmadhikari (2023) stated that the next level of policy decisions and regulations, after promoting rigorous financial inclusion, involves integrating technology to provide banking services to the mass population. One of them is the utilization of artificial intelligence. As digital financial platforms have expanded, financial institutions have increasingly integrated artificial intelligence technologies to enhance service delivery, improve risk management, and expand reach to underserved populations. Artificial intelligence is broadly defined as “the science and engineering of making intelligent machines, especially intelligent computer programs (McCarthy 2007, p. 2). It is related to a similar task using computers to understand human intelligence.” Sheikh et al. (2023) pointed out that there is no standard definition of the concept yet. De Zúñiga et al. (2024, p. 1) presented an AI scholarly definition in communication research “as the tangible real-world capability of non-human machines or artificial entities to perform, task solve, communicate, interact, and act logically as it occurs with biological humans.” Kothandapani (2022) described it as the ability of a machine to process information and make decisions or forecasts without needing specific, step-by-step programming. Bank for International Settlements (2024, p. 93) defined Artificial Intelligence

as “a broad term, referring to computer systems performing tasks that require human-like intelligence”.

Kothandapani (2022) stressed that the enhancement of computation capacity and algorithms, and data accessibility have evolved the AI applications in the financial services sector. Thakkar (2024) highlighted how AI-driven fintech solutions, including credit scoring, chatbots, and fraud detection systems are promoting financial inclusion within emerging markets. According to Yanting and Ali (2023), Artificial Intelligence (AI) has significantly enhanced credit scoring and risk assessment through the use of non-traditional data sources, facilitating access to loans for individuals with limited credit histories and low incomes. They also highlighted AI-powered customer identification and verification systems have improved security measures. Also, artificial intelligence empowers fintech and bigtech companies to convert soft information collected through social media or alternative methods into hard information (Beck 2020). Machine learning, a subset of Artificial Intelligence, can be used for evaluating credit scoring and enhancing financial inclusion (Bazarbash, 2019; Óskarsdóttir et al., 2019). How et al. (2020) acknowledged the AI complexity and so developed a research paper showcasing a no-coding human-centric AI-based approach to demonstrate AI for individuals without a strong background in computer science. According to them, this approach has the potential to simulate interactions between financial profiles and predict individuals' intentions regarding the offered financial products. For Telukdarie and Mungar (2023), artificial intelligence, as part of digital technology implementation, may serve as an effective instrument to linking individuals in developing markets to financial institutions. However, despite AI's enumerated positive effects, it has challenges and limitations, such as AI-based FinTech projects are still in the prototyping stage even in the developed countries, a shortage of AI talent, and concerns about negative impacts on employment and privacy. Nonetheless, Kshetri (2021) believed that these challenges are likely to be conquered with maturity of technology. Tan (2025) acknowledged the pivotal force of AI in Financial Technology (FinTech), which is revolutionizing analysis, automation, and decision-making. He stated that it reduces operational and labor costs and improves the speed and accuracy of financial transactions such as loan assessments, fraud detection, and investment management. However, he also pointed out the challenges of integrating AI into FinTech; concerns over algorithmic bias, lack of transparency, and the accountability of how AI systems make decisions remain as significant challenges.

2.5 Variables Used in Financial Inclusion Studies

Existing literature on financial inclusion employs a diverse yet convergent set of variables to quantify and explain inclusion outcomes. Across multiple studies, financial inclusion is commonly measured through account ownership, use of formal savings and credit services, engagement with digital financial platforms, and, in some cases, composite indices of financial inclusion or financial well-being developed using techniques such as principal component analysis (Debuque-Gonzales and Corpus 2021, 2024; Mojica and Mapa 2017; Tan 2014). Explanatory variables frequently encompass socio-demographic characteristics, which includes education, income, gender, age, and urban-rural residence, as well as institutional and structural factors such as financial literacy, access to digital infrastructure, and the availability of formal financial institutions (e.g. bank branches, ATMs).

Moreover, broader contextual variables have been incorporated in cross-country analyses. For example, Anyangwe, Vanroose, and Fanta (2024) integrate cultural dimensions derived from Hofstede's framework, while studies such as those by Debuque-Gonzales and Corpus (2021, 2024), and Nandru, Chendragiri, and Velayutham (2021) emphasize digital access and

technological readiness as critical determinants of inclusion. Despite variations in context and methodology, the empirical evidence consistently indicated that education, income, and financial literacy are the most significant predictors of financial inclusion. Conversely, gender, regional, and socio-economic disparities persist as barriers, particularly in developing economies. Collectively, these studies underscore that financial inclusion is multidimensional driven by both access-related factors (e.g. infrastructure, digital connectivity) and capability-related factors (e.g. education, literacy, and income). A summary of all these studies is presented in Appendix A.

2.6 Existing Policy Frameworks

The government's commitment to leveraging AI and digital innovation to strengthen the financial sector is reflected in several key policy frameworks:

National Strategy for Financial Inclusion (NSFI) 2022–2028. The NSFI serves as the Philippines' comprehensive roadmap for expanding financial inclusion. Although it does not explicitly focus on AI, it highlights the central role of digital technologies in lowering service delivery costs, promoting innovation, and reaching underserved populations. One of its core objectives is the promotion of inclusive digital finance.

National AI Strategy Roadmap 2.0 (2024). Launched by the Department of Trade and Industry, the updated AI roadmap outlines the country's vision to become a regional hub for AI. It identifies priority sectors—including financial services—where AI can enhance productivity, improve operational efficiency, and support innovation. The roadmap also emphasizes capacity-building and the development of supportive research and regulatory ecosystems.

BSP Digital Payments Transformation Roadmap (2020–2023). This roadmap sets the Bangko Sentral ng Pilipinas' strategic direction for accelerating the adoption of digital payments and expanding account ownership. It targets a significant shift toward digital retail payments and broader participation in the formal financial system, laying essential groundwork for AI-enabled financial services.

BSP Project Sapiens (2024). Project Sapiens is BSP's thematic review of AI and machine learning applications in the Philippine financial sector. It identifies common use cases—such as e-KYC, fraud detection, credit scoring, and recommender systems—as well as emerging risks and varying levels of institutional readiness. The BSP underscores the need for proportionate AI governance frameworks tailored to each institution's risk profile.

Data Privacy Act of 2012. The Data Privacy Act provides the legal foundation for responsible data processing in digital and AI-enabled financial services. Its principles on transparency, fairness, and data protection are central to ensuring that AI applications, such as automated credit scoring and fraud management, operate in a safe and ethical manner.

National Cybersecurity Plan 2023–2028. The National Cybersecurity Plan strengthens the country's defense against cyber threats and supports a secure environment for digital financial transactions. It reinforces the importance of resilient digital infrastructure, which is a prerequisite for safe AI adoption in the financial sector.

Republic Act 12234 (Konektadong Pinoy Act). This law establishes a national framework to expand digital connectivity, aiming to reduce the digital divide and ensure broader access to

online services. Improved connectivity supports both digital financial inclusion and the data ecosystems required for AI-driven financial services.

Republic Act 12254 (E-Governance Act). This law institutionalizes the transition to interoperable and efficient digital public services across government. By promoting digitization of records and standardized ICT systems, it expands the availability of digital data and enhances the enabling environment for AI innovations, including in financial applications.

DOST–PCIEERD ICT and AI Roadmap (2020). This roadmap outlines national priorities for AI and ICT research, focusing on innovation, automation, and data-driven development. While broad in scope, it reinforces the country’s long-term commitment to building AI capabilities that can also benefit the financial sector.

As AI continues to develop due to advances in technology, policy frameworks are also continuously refined to match the challenges that these innovations present especially in the financial sector. Through collaborations with different stakeholders and with the support of the government, inadequacies are aimed to be resolved, thus maximizing the potential of AI in increasing financial inclusion in the Philippines.

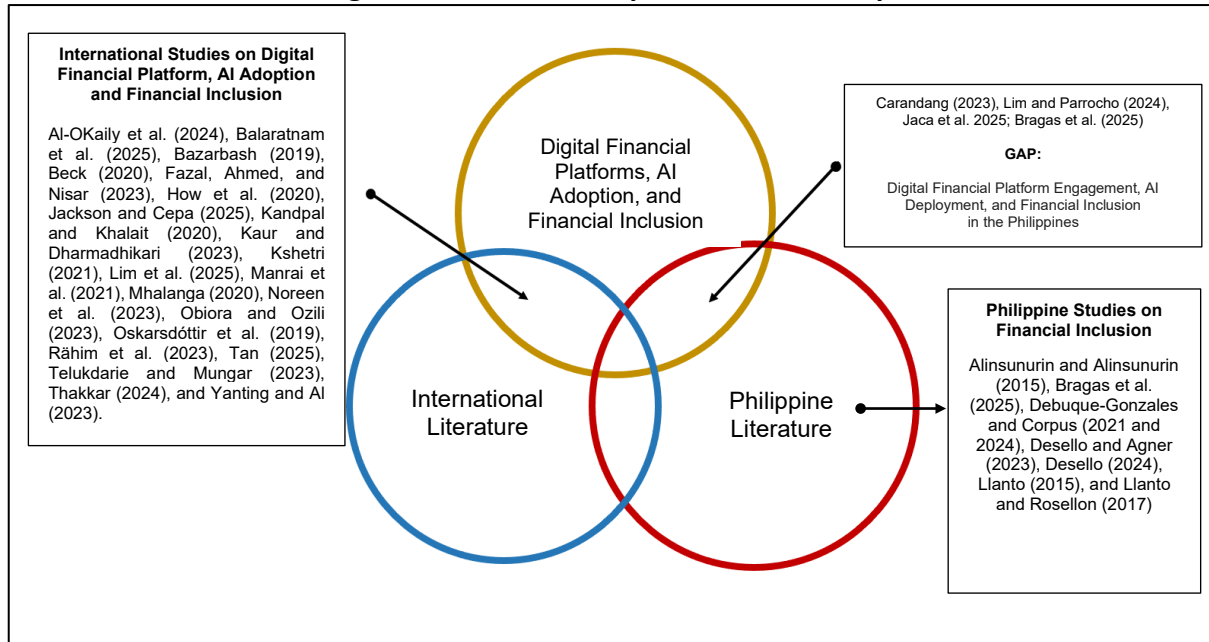
2.7 Research Gap

Philippine studies on financial inclusion, utilizing nationwide data, have predominantly focused on examining its determinants, such as demographic and social variables (Llanto 2015; Llanto and Rosellon 2017; Alinsunurin and Alinsunurin 2015), measuring financial inclusion indices (Debuque-Gonzales and Corpus 2021), and examining the effect of financial literacy on financial inclusion (Desello and Agner 2023). More recently, Debuque-Gonzales and Corpus (2024) examined disparities in the ownership and use of financial technology, particularly mobile money, across education and income levels, highlighting emerging digital divides in financial access. While these studies provide valuable insights on digital financial access, what remains less explored is a comprehensive examination of digital financial platform engagement, encompassing not only mobile money but also online banking, digital payments, and e-commerce transactions, and how this broader pattern of digital behavior relates to financial inclusion outcomes. Moreover, research on artificial intelligence in the Philippine financial sector has primarily examined AI from the supply side, focusing on institutional adoption and customer perceptions (Carandang 2023; Lim and Parrocho 2024; Jaca et al. 2025; Bragas et al. 2025), without connecting these developments to consumer engagement patterns.

This study contributes by taking a two-sided approach that bridges these perspectives. On the demand side, a comprehensive Digital Financial Engagement Index is constructed to capture multiple dimensions of digital financial behavior, including mobile phone ownership, internet access, mobile payments, online banking, digital transactions, and e-commerce, and the relationship between this engagement and financial inclusion outcomes are examined. On the supply side, the deployment of AI technologies within digital financial platforms by financial institutions is documented using institutional data and key informant interviews. By connecting consumer behavior with institutional capacity, the study provides a more complete picture of how digital financial platforms, increasingly enhanced by AI technologies, can advance financial inclusion in the Philippines. This integrated approach offers insights for policymakers on strengthening digital financial infrastructure, promoting digital literacy, and ensuring responsible AI deployment to advance inclusive financial services. This is particularly relevant

as digital transactions become mainstream and financial institutions increasingly integrate AI into their operations. Figure 4 shows the literature map emphasizing the research gap.

Figure 4. Literature Map and Research Gap



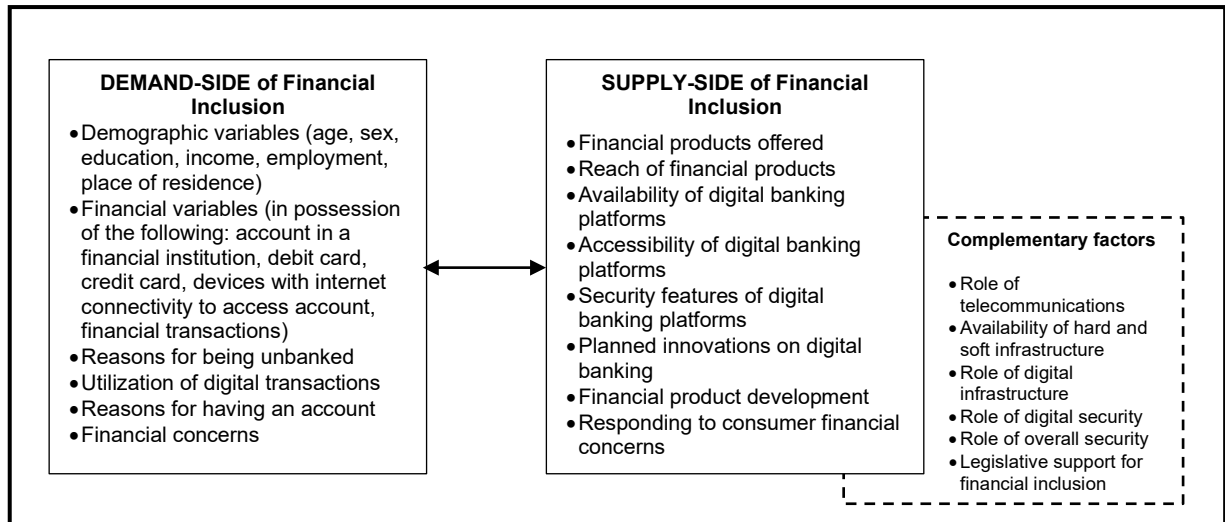
Source: Constructed by the authors

3. Framework, Research Design and Methodology

3.1 Research Framework

In addressing the problem statement and research objectives, a market approach is employed looking into both demand- and supply-side facets of financial inclusion. This probes on what the consumers prefer (demand-side) and what financial institutions can feasibly provide to satisfy the needs of the consumers (supply-side). The conceptual framework is illustrated in Figure 5.

Figure 5. Conceptual Framework on Digital Financial Engagement and Financial Inclusion

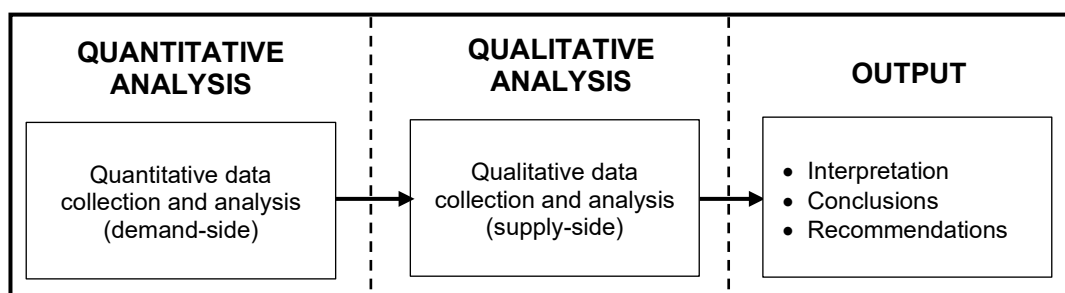


Source: Constructed by the authors.

3.2 Research Design

From the framework presented in Figure 5, the study employed a qualitative analysis to address the problem statement and research objectives. An explanatory sequential mixed method is employed as seen in Figure 6. The research team used a quantitative method to analyze the demand-side, followed by qualitative key informant interviews for the supply-side. Following this order, the quantitative method is used first, and the qualitative method is used to explore unexpected results. Hence, the qualitative findings not only enrich and contextualize the quantitative results but also bridge any gaps between the two perspectives, ensuring a comprehensive understanding of AI's role in advancing financial inclusion. Afterwards, these results were utilized to make interpretations, conclusions, and recommendations.

Figure 6. Explanatory Sequential Mixed Method



Source: Constructed by the authors.

3.3 Quantitative Analysis (Demand-Side)

To examine how digital financial platform engagement influences financial inclusion from the demand-side perspective, the study made use of the Global Financial Inclusion (Global Findex) database 2021 for the Philippines. Given that the Global Findex does not include direct measures of AI awareness, knowledge, or usage, digital financial platform engagement is used as a behavioral proxy for individuals' potential exposure or interaction to AI-enabled financial services. This approach recognizes that AI is embedded unevenly across digital financial platforms and is typically encountered through the use of mobile payments, online banking, and digital transactions rather than through directly observable AI usage. The Global Findex provides the world's most complete dataset on financial inclusion as it provides demand-side data for global and regional cross-country analysis, which enables researchers to create a thorough and multidimensional picture of individuals' behaviour regarding saving, borrowing, making payments, and managing risks (World Bank 2022). The survey data covers 128,000 respondents in 123 economies, capturing 91 percent of the global population. The data has been published every three years since 2011. It has 1,000 cases with 114 variables. Since the Global Findex is cross-sectional with a number of nominal-level variables, the data will be subjected to cross tabulations and contingency table analysis.

3.3.1 Construction of the Digital Financial Engagement Index

Measuring individual-level interaction to artificial intelligence (AI) is challenging because large household surveys, including the Global Findex, do not capture direct AI use, awareness, or literacy (World Bank 2021). Moreover, the extent of AI integration varies significantly across digital financial platforms, with some extensively utilizing AI for functions such as fraud detection and credit scoring, while others rely more on traditional rule-based algorithms. These platform-level differences are not observable in survey data that simply asks whether respondents use digital payments.

Given these limitations, this study constructs a Digital Financial Engagement Index that measures individuals' usage of and interaction with digital financial platforms and services. Digital financial platforms constitute the primary delivery channels through which consumers encounter AI-enabled financial services, even when users are unaware of the underlying technology (Manta et al. 2025; Deloitte 2025; McKinsey 2025). The index should therefore be interpreted as measuring digital financial behavior and platform engagement, the behavioral foundation that enables potential exposure to AI-enhanced services, rather than direct AI literacy or awareness. The connection to AI is established through the convergence of demand-side behavior, reflected in platform usage, and supply-side deployment, which documents which platforms actually incorporate AI, as discussed in Section 4.2.2.

Building on this framework, the index uses proxy indicators of digital behavior to estimate environments in which individuals are more likely to encounter AI-enabled financial services. Contemporary financial platforms, such as mobile banking applications, electronic wallets, and online payment systems, increasingly integrate AI functions in back-end operations, including automated fraud detection, credit scoring algorithms, recommender systems, and customer service chatbots (Manta et al. 2025; Deloitte 2025). As a result, routine digital behaviors provide a useful indication of exposure to the digital financial ecosystem within which AI is being deployed, even though they do not measure AI interaction directly (McKinsey 2025). Accordingly, the measure should be interpreted as capturing the behavioral channels through which AI-enabled services are delivered, with the understanding that actual AI exposure varies

depending on which specific platforms individuals use and how extensively those platforms have integrated AI technologies. Future surveys that directly measure AI awareness and platform-specific usage could provide more granular insights into heterogeneity in AI exposure (Felten et al. 2021).

The construction of the index involved the following steps. First, eight binary variables were selected as proxies for digital financial engagement, covering aspects of digital access and financial behavior: mobile phone ownership, internet access, making any digital payment, using a mobile phone to pay for in-store purchases, paying bills online, sending money online to a relative or friend, making online purchases, and making digital merchant payments. These variables were chosen for their potential to reflect an individual's engagement with technologies through which AI applications are commonly delivered (e.g., e-wallets, online banking platforms, and mobile apps). Before proceeding with dimensionality reduction, internal consistency was assessed using Cronbach's alpha to determine whether the variables collectively capture a coherent construct related to Digital financial engagement. The analysis yielded a Cronbach's alpha of 0.82, based on eight (8) binary items and an average inter-item covariance of 0.064, indicating good internal reliability.

Second, the selected variables were standardized to ensure comparability. This step is necessary because the binary indicators vary in prevalence and may otherwise contribute unevenly to the underlying dimensions in the analysis. Standardization balances the influence of each variable and improves the interpretability of the results. Third, Multiple Correspondence Analysis (MCA) was applied to the standardized dataset to extract latent dimensions of digital behavior and access. MCA is a dimensionality reduction technique appropriate for categorical data, commonly used to derive composite indices from sets of binary variables. The first principal dimension, which captures the greatest proportion of explained variance, was retained and used as the basis for constructing the Digital Financial Engagement Index. Finally, the index values were normalized to a 0–1 scale, where higher scores indicate greater engagement with digital financial platforms and services.

3.3.2 Regression Analysis

The regression models examine the relationship between the Digital Financial Engagement Index and two key dimensions of financial inclusion, which serve as the dependent variables, following the framework of (Debuque-Gonzales and Corpus 2021). The first dimension, account ownership, includes indicators such as having a formal financial account, owning a debit card associated with a financial account, having a mobile money account, and holding a credit card. The second dimension, account usage, refers to behaviours such as formal saving, formal borrowing, transactions on remittances through a financial institution, transacting via a mobile phone, and online financial transactions using the internet.

Each of these dependent variables is binary (1 = yes, 0 = no) and modeled using logistic regression. The general form of the model for account ownership and usage is:

$$\text{logit} (P(Y_i = 1)) = \beta_0 + \beta_1 DFEI_i + \beta_2 X_i + \varepsilon_i$$

where Y_i represents the financial inclusion outcome (e.g., owning a debit card or saving formally), $DFEI_i$ is the Digital Financial Engagement Index, and X_i is a vector of control variables including age group (categorized), gender (1 = male, 0 = female), educational attainment (ordinal), income quintile, employment status (in or out of the labor force), and

urbanicity (urban or rural). The coefficients β_0 and β_1 (where $i = 1, 2$) represent the model intercept and the marginal effects of each explanatory variable, respectively, while ε_i is the error term. Sample weights were applied to ensure the estimates reflect nationally representative patterns.

In addition to the financial inclusion outcomes, a complementary set of models examines perceived barriers to account ownership among unbanked individuals. While not a direct measure of inclusion, these self-reported constraints such as lack of money, high cost, distance to institutions, lack of documentation, religious reasons, or distrust offer crucial insights into involuntary exclusion. Each barrier is modeled as a binary dependent variable using the same set of explanatory variables as the financial inclusion models. The corresponding equation is:

$$\text{logit}(P(B_i = 1)) = \gamma_0 + \gamma_1 DFEI_i + \gamma_2 X_i + v_i$$

where B_i is a specific self-reported barrier to account ownership. The coefficients γ_0 and γ_1 (where $i = 1, 2$) represent the intercept and marginal effects of the explanatory variables, respectively, while v_i is the error term. Sample weights were again applied to ensure the estimates reflect nationally representative patterns. This analysis supplements the inclusion models by addressing not just who is included and what services are used, but also why certain individuals remain outside the formal financial system. Together, these models examine the determinants of financial inclusion (account ownership and usage) and the barriers that prevent account ownership among the unbanked.

Before running the regression models, descriptive statistics were generated to explore key patterns in financial inclusion and barriers to account ownership across socio-economic groups. Appendix D shows the average rates of formal financial account ownership, including debit and credit card ownership and mobile money use. Appendix E presents the average rates of formal financial service usage, such as saving, borrowing, and making digital transactions. Appendix F summarizes the most commonly reported barriers to account ownership. These descriptive tables help highlight existing disparities and provide important context for interpreting the regression results.

3.4 Qualitative Analysis (Supply-Side)

To provide a broader understanding of the supply-side enablers and constraints related to financial inclusion, this study utilizes time-series data from the IMF Financial Access Survey (FAS) for the Philippines covering the period 2016 to 2024. The FAS provides detailed indicators on the country's financial access infrastructure and institutional capacity, including trends in access points, deposit and credit services, and regulatory frameworks. These data help contextualize the evolution of the country's financial ecosystem and support an assessment of its readiness to integrate AI-enhanced financial services.

To complement this institutional data, the study also employs key informant interviews (KIIs) with selected policymakers and industry experts. KIIs involve a select group of stakeholders and individuals that are chosen given the experience, information, and insights they have on the subject at hand (Kumar 1989). Key informants are deemed expert sources of information (Marshall, 1996) and can expand our knowledge and understanding, including nuances and peculiarities, about a specific subject-matter (Cossham and Johanson 2019). The selected key informants include experts from organizations involved in financial inclusion such as academe, public policy, financial institutions (e.g. banks, cooperatives, associations), and data science.

A total of 20 experts were invited to meet the 12 confirmed informants. A structured in-depth interview was implemented wherein the questions posed to the participants included topics listed in Table 4.

Table 4. Topics of KII questionnaire

Financial Inclusion Metrics	Topics
Market access and development: Access to traditional financial products	• Reasons of consumers for being banked and unbanked • Knowledge on traditional financial products and services • Availability and access to traditional financial products
Market access and development: Access to digital financial products	• Reasons of consumers for being banked and unbanked • Knowledge on traditional financial products and services • Availability and access to traditional financial products
Reasons of consumers for being banked and unbanked	• Market development and promotion • Concerns on access, availability, and connectivity • Concerns on safety and security
Technology and Technology Sharing	• Adoption of digital transaction platforms • Bandwagon effect of digitalization in the financial sector • Information on best practices on the use of ICT and AI
Access to Information and Communication Technology (ICT)	• Information on financial products • Information of government regulations and policies on financial products • Information on government interventions in the financial sector • Possible AI interventions towards financial inclusion
Policy Environment	• Presence of public-private sector dialogue • Presence/lack of policy support • Other legislative support required to enhance financial inclusion
Note: For the list of key informant interview questions and key informant interview consent forms, please refer to Appendix G and H, respectively. Source: Constructed by Authors	

3.5 Validation

The study employed a triangulation method as a technique to cross verify data from multiple sources (Johnson et al. 2017). For the purposes of this study, the data from the demand-side analysis (Global Findex survey), supply-side analysis (IMF Financial Access Survey and key informant interviews), and literature review are synthesized to provide a comprehensive assessment of financial inclusion and AI deployment in the Philippine financial sector. The triangulation process involves comparing patterns observed in consumer behavior (demand side) with institutional capacity and AI adoption patterns (supply side). Quantitative findings on digital financial engagement and financial inclusion are validated against qualitative insights from key informant interviews with financial sector experts. Findings are also assessed for consistency with established literature on technology adoption and financial inclusion.

4. Results of the Study

This section presents a comprehensive overview of financial inclusion in the Philippines by examining both demand-side and supply-side indicators related to access, usage, and infrastructure. Drawing from the 2021 World Bank Global Findex and the 2016–2024 IMF Financial Access Survey (FAS), the analysis highlights key behavioral patterns and

institutional trends, identifying gaps and opportunities for promoting broader and more inclusive financial participation.

4.1 Demand-Side Analysis (World Bank Global Findex 2021)

4.1.1 Account Ownership and Usage Patterns

Table 5 presents key indicators of account ownership and usage among Filipino adults based on World Bank Findex 2021 data. Slightly more than half of respondents (56%) reported owning a formal financial account, while 39% had a debit card and 24% had a credit card (among those with accounts). Ownership of mobile money accounts was comparatively lower at 30%, suggesting that adoption of mobile-based financial services remains in its early stages.

In terms of account usage, only 28% reported saving and 16% reported borrowing through formal institutions. Among cardholders, 49% used their debit cards and 77% use their credit cards, indicating higher engagement among credit users. In addition, 27% of adults reported using formal remittance channels, while 39% engaged in mobile-based financial transactions and 56% made transactions online. These figures point to uneven patterns of financial engagement, where digital channels show potential for broader inclusion, even as the use of formal credit and savings services remains limited.

Table 5. Financial Inclusion Indicators – Account Ownership and Usage

Category	Variable	Obs	Mean	Std Dev.
Account Ownership	Formal financial account	1,000	0.56	0.50
	Debit card	1,000	0.39	0.49
	Credit card	496	0.24	0.43
	Mobile money account	496	0.30	0.46
Account Usage	Formal savings	1,000	0.28	0.45
	Formal borrowing	1,000	0.16	0.36
	Debit card usage	388	0.49	0.50
	Credit card usage	119	0.77	0.42
	Formal remittance	1,000	0.27	0.44
	Mobile usage	1,000	0.39	0.49
	Online usage	1,000	0.56	0.50

Source: World Bank Findex 2021; Authors' estimates.

Notes: Credit card ownership is based on respondents with formal financial accounts. Debit and credit card usage rates are based on cardholders only. All variables recoded to binary (1 = yes, 0 = no).

4.1.2 Barriers to Financial Inclusion

Table 6 presents the main reasons cited by Filipino adults for not owning a formal financial account. The most commonly reported barrier is lack of money (76%), suggesting that affordability remains a critical issue in financial inclusion. This is followed by perceptions of high cost (55%) and physical distance from financial institutions (40%), highlighting both economic and geographic constraints. Documentation issues (39%) and a perceived lack of need for financial services (37%) also emerged as significant impediments. Trust in financial institutions is relatively low, with 20% citing it as a reason, while religious concerns (13%) were the least cited. Notably, 34% said they did not need an account because a family member already had one, suggesting shared use within households. These findings emphasize the need

to reduce financial and logistical barriers while building trust and ensuring that financial services met the needs of low-income and underserved populations.

Table 6. Reasons for not having a formal account

Variable	Obs	Mean	Std Dev.
Too far	504	0.4	0.49
Too expensive	504	0.55	0.52
Lack of documentation	504	0.39	0.49
Lack of trust	504	0.2	0.4
Religious reasons	504	0.13	0.34
Lack of money	504	0.76	0.43
Family already has one	504	0.34	0.5
No need for financial services	504	0.37	0.48

Source: World Bank Findex 2021; Authors' estimates

Note: All variables recoded to binary (1 = yes, 0 = no).

Table 7. Formal Saving and Borrowing Behavior Among Filipino Adults

Variable	Obs	Mean	Std. Dev.
Borrowed from a financial institution	1,000	0.158	0.365
Saved using an account at a fin. institution	1,000	0.282	0.45
General borrowing in past year (borrowed)	1,000	0.612	0.488
General saving in past year (saved)	1,000	0.645	0.479

Source: World Bank Findex 2021; Authors' estimates.

Note: All variables recoded to binary (1 = yes, 0 = no).

Table 7 shows that access to formal financial services remains limited for many Filipinos. Only 15.8% of respondents reported borrowing from a financial institution, and 28.2% reported saving using an account at a formal institution. However, the overall rates of borrowing and saving were significantly higher, at 61.2% and 64.5%, respectively. This gap suggests that a large share of financial activity takes place outside the formal sector.

As shown in Table 8, informal borrowing remains widespread: 42.8% of respondents borrowed from family or friends, and among a smaller group of 95 respondents, 48.4% borrowed from an informal savings club. In addition, 9.5% of respondents reported saving through informal savings clubs. These figures indicate that informal financial mechanisms continue to play a central role in the financial lives of many Filipinos. This reliance may reflect several persistent barriers to formal access, such as cost, documentation requirements, or lack of trust (Debuque-Gonzales and Corpus 2021). Informal practices are especially important in rural or low-income settings, where formal infrastructure is often limited. As such, strategies to improve financial inclusion should not only expand formal services but also recognize and build upon existing informal practices that remain relevant and trusted by underserved communities.

Table 8. Informal Saving and Borrowing Behavior Among Filipino Adults

Variable	Obs	Mean	Std. Dev.
Borrowed from family or friends	1,000	0.428	0.495
Borrowed from an informal savings club	95	0.484	0.502
Saved using an informal savings club	1,000	0.095	0.285

Source: World Bank Findex 2021; Authors' estimates.

Note: Sample size for informal borrowing club limited to those who responded.

Table 9 presents summary statistics for the variables used to construct the Digital Financial Engagement Index in the Philippines, covering indicators of digital access (such as mobile phone ownership and internet use) and digital financial behavior (such as online payments and money transfers). The data show high levels of mobile phone ownership (96.3%) and internet access (87.4%), indicating strong digital connectivity. However, the use of digital financial services remains moderate. Only 54.7% of respondents made any digital payment, with lower uptake for specific activities such as online bill payments (30.2%) and digital merchant payments (26.3%). This suggests that while digital access is widespread, active engagement with digital finance is still evolving.

Table 9. Key Indicators of Digital Access and Financial Behavior for the Digital Financial Engagement Index

Definition	Mean	Std. Dev.
Owns a mobile phone	0.963	0.189
Has access to the internet	0.874	0.332
Made any digital payment in the past year	0.547	0.498
Used mobile phone to pay for in-store purchase	0.177	0.382
Paid bills online using the internet	0.302	0.459
Sent money online to a friend or relative using the internet	0.351	0.478
Made a digital merchant payment (e.g., QR code, e-wallet)	0.263	0.440

Source: World Bank Findex 2021; Authors' estimates.

Note: All variables are binary (1 = yes, 0 = no) and based on 1,000 observations.

4.1.3 Construction of the Digital Financial Engagement Index

To construct a Digital Financial Engagement Index, a Multiple Correspondence Analysis (MCA) was performed on seven categorical variables capturing ownership, access, and usage of digital and financial technologies namely, mobile phone ownership, internet access, use of any digital payment method, and engagement in specific financial transactions (bill payments, remittances, merchant payments, etc.).

Two dimensions were extracted, explaining 94.1% of the total inertia (Dim 1 = 93.9%, Dim 2 = 0.17%). Since the first dimension accounted for nearly all the variance, it was retained as the basis for the Digital Financial Engagement Index. Dimension 1 reflects a gradient of digital financial engagement, ranging from individuals with limited access to enabling technologies (no mobile phone, no internet, no digital payments) to those who frequently conduct financial transactions using digital platforms. The largest positive contributors were participation in online bill payment (fin14_1 = yes, 12.3%), remittances (fin14a = yes, 12.9%), merchant payments (merchantpay_dig = yes, 14.4%), and other financial activities (fin14a1 = yes, 9.6%; fin14b = yes, 7.5%). Negative contributions corresponded to lack of access and non-use of digital tools (internetaccess = no, mobileowner = no, anydigpayment = no). The second dimension captured only 0.17% of inertia and exhibited no clear thematic clustering, hence it was not used in index construction.

The row scores for the first dimension were normalized to a 0–1 scale to create the Digital Financial Engagement Index, where higher values indicate greater engagement with digital financial technologies. The index ranged from 0 to 1 (mean = 0.44, SD = 0.30), suggesting moderate heterogeneity in digital financial engagement across respondents. This index later served as a key explanatory variable in the empirical analysis linking Digital Financial Engagement to financial inclusion outcomes. Detailed results of the Multiple Correspondence

Analysis (MCA), including category coordinates and contributions to the retained dimension, are presented in Appendix B. Summary statistics of the Digital Financial Engagement Index and its key indicators, disaggregated by socio-economic group, are provided in Appendix C.

4.1.4 Regression Analyses

4.1.4.1 Determinants of Account Ownership

Table 10 presents the marginal effects from the logit regressions on various forms of account ownership: traditional financial accounts, debit cards, credit cards, and mobile money accounts. The estimations use survey weights and standard errors are reported in parentheses.

As hypothesized, the Digital Financial Engagement Index is a strong and consistent predictor of account ownership across all types. A one-unit increase in Digital Financial Engagement Index is associated with a 78.5 percentage point increase in the likelihood of owning a formal account, 48.3 points for debit card ownership, 41.0 points for credit card ownership, and 59.3 points for mobile money ownership – all statistically significant at the 1% level. This highlights the enabling role of digital financial platform engagement in financial access.

In terms of age, older adults (35–44) are significantly more likely to own formal and debit card accounts, while seniors (60+) are especially more likely to own credit cards (by 29.3 percentage points). Interestingly, mobile money ownership does not significantly differ across age groups, and seniors are slightly less likely to use it.

The effect of gender is not statistically significant for any ownership type, suggesting a diminishing gender gap in financial access.

Education plays a more prominent role in traditional account ownership. Individuals with tertiary education are 16.5 percentage points more likely to own a formal account and 18.4 points more likely to own a debit card compared to those with only primary education. However, education does not significantly affect mobile money or credit card ownership in this sample.

Across income quintiles, those in the 5th (richest) quintile are more likely to own debit cards (13.4 points) and credit cards (22.5 points) than the poorest quintile. However, no strong pattern emerges for mobile money ownership, reinforcing the idea that such services may be more accessible to lower-income groups, or that usage is still limited regardless of income.

Finally, employment status shows that those in the labor force are more likely to own credit cards (10.2 percentage points) compared to those out of the workforce. The effect is modest or insignificant for other account types.

Table 10. Determinants of Account Ownership

	(1)	(2)	(3)	(4)
VARIABLES	Formal Account	Debit Card	Credit Card	Mobile Money
Digital Financial Engagement Index	0.785*** (0.076)	0.483*** (0.056)	0.410*** (0.069)	0.593*** (0.043)
<i>Age group (Youth, 15–24, ref.)</i>				
Young adults (25–34)	0.088* (0.047)	0.073* (0.038)	0.092* (0.054)	0.087*** (0.028)
Mid adults (35–44)	0.135*** (0.050)	0.137*** (0.045)	0.034 (0.062)	0.025 (0.029)
Older adults (45–59)	0.086* (0.050)	0.096** (0.046)	0.093 (0.078)	0.022 (0.040)
Senior (60+)	0.186** (0.081)	0.218*** (0.081)	0.293*** (0.094)	-0.047 (0.068)
<i>Gender (Male, ref.)</i>				
Female	0.005 (0.036)	0.022 (0.030)	-0.004 (0.044)	-0.005 (0.023)
<i>Education (Completed primary school or less, ref.)</i>				
Completed secondary school	0.054 (0.043)	0.046 (0.042)	-0.044 (0.108)	-0.019 (0.030)
Completed tertiary education or more	0.165** (0.070)	0.184*** (0.061)	-0.052 (0.102)	-0.041 (0.038)
<i>Income quintiles (1st income quintile)</i>				
2 nd income quintile	-0.005 (0.056)	-0.041 (0.059)	0.074 (0.121)	0.034 (0.046)
3 rd income quintile	-0.013 (0.056)	-0.026 (0.054)	0.12 (0.076)	0.013 (0.035)
4 th income quintile	0.077 (0.057)	0.025 (0.055)	0.109* (0.064)	0.023 (0.037)
5 th income quintile	0.099 (0.064)	0.134** (0.065)	0.225*** (0.077)	0.029 (0.040)
<i>Employment (Out of workforce)</i>				
In the labor force	0.055 (0.039)	0.048 (0.035)	0.102** (0.046)	-0.018 (0.026)
Observations	1,000	1,000	496	1,000

Note: Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

4.1.4.2 Determinants of Account Usage

Table 11 presents the marginal effects from logit regressions examining the likelihood of using accounts for various financial purposes, including formal saving, borrowing, debit and credit card usage, remittance transactions, and fintech-based usage via mobile and internet platforms. Each column corresponds to a distinct financial behavior, and the marginal effects reflect percentage point changes in the predicted probability of account usage, holding other factors constant.

Digital financial engagement emerges as a consistently strong and statistically significant predictor across all forms of account usage. For example, a one-unit increase in the Digital

Financial Engagement Index is associated with an increase of 34.5 percentage points in the likelihood of formal saving (Column 1), 24.4 points for formal borrowing (Column 2), and as high as 157.0 percentage points for online usage (Column 7), suggesting that digital financial engagement is a key enabler of financial inclusion via both traditional and digital channels.

The relationship between age group and account usage is generally nonlinear. Compared to youth (ages 15–24), older adults (ages 45–59) are significantly more likely to borrow formally (8.8 percentage points), use credit cards (26.4 percentage points) and remit formally (10.4 percentage points). However, they are significantly less likely to use online platforms for financial transactions (–8.5 points). A similar negative relationship with online usage is observed across all older age groups (ages 25 and above) when compared to youth aged 15–24, underscoring a digital divide by age.

Gender does not appear to significantly predict any type of account usage in this model, suggesting that the gender gap in account usage might be narrowing or is influenced by other intersecting factors like income and education.

In terms of educational attainment, individuals with tertiary education are 8.6 percentage points more likely to save formally and 35.8 points more likely to use credit cards than those with primary or less schooling. Interestingly, education does not significantly influence mobile or online usage in this model, which may indicate that mobile platforms are becoming more accessible regardless of formal education.

Income quintiles show a clear pattern, particularly for debit card use, where individuals in the 2nd income quintile are 54.5 percentage points more likely to use a debit card than those in the poorest quintile. Similarly, those in the 3rd and 5th quintiles are significantly more likely to save using an account at a financial institution. Online usage is lower among the poorest quintile, with the 5th quintile being 7.6 points less likely than the richest, a somewhat surprising inverse pattern which warrants further exploration.

Employment status is significantly associated with higher likelihood of formal saving and borrowing. Being in the labor force is associated with a 6.1 and 4.6 percentage point increase in saving and borrowing, respectively, compared to those outside the workforce. However, employment status does not significantly predict usage of fintech channels such as mobile or online platforms.

In summary, these results highlight that digital financial engagement is a key determinant of financial account usage, particularly for fintech-related services. Alongside education and income, digital financial engagement significantly influences individuals' ability to engage with formal financial systems. The strong and consistent relationship between digital financial engagement and usage outcomes suggest that promoting digital financial platform adoption can help expand financial inclusion, especially among older and lower-income populations. These findings offer important guidance for supply-side strategies aimed at improving the reach and effectiveness of digital financial services.

Table 11. Determinants of Account Usage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
VARIABLES	Formal Savings	Formal Borrowing	Debit Card Use	Credit Card Use	Formal Remittance	Mobile Usage	Online Usage
Digital Financial Engagement Index	0.345*** (0.040)	0.244*** (0.049)	0.636*** (0.080)	0.716*** (0.136)	0.530*** (0.042)	0.793*** (0.056)	1.570*** (0.076)
<i>Age group (Youth, 15–24, ref.)</i>							
Young adults (25–34)	0.022 (0.027)	0.028 (0.031)	0.056 (0.084)	0.182 (0.110)	0.028 (0.036)	0.028 (0.028)	- 0.083*** (0.028)
Mid adults (35–44)	0.026 (0.033)	0.073* (0.040)	0.059 (0.074)	-0.019 (0.129)	0.068* (0.039)	0.006 (0.030)	-0.061* (0.036)
Older adults (45–59)	0.004 (0.036)	0.088* (0.046)	0.049 (0.100)	0.264** (0.125)	0.104** (0.044)	0.03 (0.036)	-0.085** (0.038)
Senior (60+)	0.03 (0.073)	-0.006 (0.046)	0.082 (0.144)	0.087 (0.081)	0.055 (0.055)	0.087 (0.081)	-0.076 (0.058)
<i>Gender (Male, ref.)</i>							
Female	0.014 (0.021)	-0.005 (0.026)	-0.003 (0.057)	0.022 (0.087)	-0.003 (0.027)	-0.034 (0.024)	0.011 (0.025)
<i>Education (Completed primary school or less, ref)</i>							
Completed secondary school	-0.002 (0.036)	-0.021 (0.036)	-0.094 (0.107)	0.346* (0.185)	0.009 (0.038)	0.001 (0.030)	0.02 (0.028)
Completed tertiary education or more	0.086* (0.049)	0.007 (0.047)	-0.094 (0.108)	0.358* (0.180)	-0.004 (0.045)	0.01 (0.039)	0.036 (0.047)
<i>Income quintiles (1st income quintile)</i>							
2 nd income quintile	0.068 (0.048)	0.022 (0.050)	0.545*** (0.202)	0 (0.048)	0.008 (0.056)	0 (0.048)	0.039 (0.033)
3 rd income quintile	0.096** (0.044)	0.023 (0.048)	0.340* (0.190)	0.069 (0.193)	-0.03 (0.048)	0.006 (0.035)	0.061* (0.034)
4 th income quintile	0.041 (0.036)	-0.018 (0.050)	0.165 (0.195)	0.156 (0.180)	-0.009 (0.051)	0.034 (0.038)	-0.02 (0.031)
5 th income quintile	0.211*** (0.053)	-0.059 (0.046)	0.275 (0.185)	0.095 (0.181)	0.023 (0.051)	0.035 (0.038)	-0.076* (0.042)
<i>Employment (Out of workforce)</i>							
In the labor force	0.061*** (0.023)	0.046* (0.027)	0.067 (0.091)	-0.1 (0.078)	-0.015 (0.032)	0.016 (0.025)	-0.006 (0.026)
Observations	1,000	1,000	1,000	1,000	1,000	1,000	1,000

Note: Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.1

Source: Authors' estimates

4.1.4.3 Determinants of Barriers to Financial Inclusion

Table 12 presents the probit regression results examining the individual-level characteristics associated with perceived barriers to account ownership. Each column reflects a different reason cited by the unbanked for not having a formal financial account. These reasons include

issues of access, affordability, documentation, trust, religion, financial constraints, household-level decisions, and lack of perceived need.

Digital financial engagement, this study's variable of interest, shows a statistically significant negative association with the probability of citing lack of trust in financial institutions as a barrier (−29.4 percentage points). This implies that individuals with greater digital financial engagement tend to have more trust in financial systems. On the other hand, digital financial engagement is positively and significantly associated with the likelihood of stating that someone in the household already has an account (+35.5 percentage points), suggesting that those with greater digital financial engagement may defer account ownership due to shared household usage rather than lack of access.

Age-related patterns in the results also reflect a digital divide. Compared to youth (ages 15–24), mid-aged adults (35–44) are significantly less likely to cite lack of documentation (−16.1 percentage points) and household-level saturation (−15.8 percentage points) as reasons for being unbanked. Interestingly, older adults (45–59) are more likely to cite religion as a barrier (+13 percentage points), indicating age-specific cultural or personal attitudes that may influence exclusion. Seniors (60+) are notably less likely to cite lack of money as a reason (−27.7 percentage points), potentially reflecting retirement incomes or reliance on household members.

Gender differences are also evident. While most gender-related coefficients are statistically insignificant, females are more likely than males to say that someone in the family already has an account (+13.1 percentage points), suggesting possible intra-household financial arrangements where women rely on male relatives' accounts.

Education does not strongly predict most barriers, although the most educated are somewhat more likely to cite lack of trust (+14.5 percentage points) and a perceived lack of need (+15 percentage points) than the least educated, possibly indicating a preference for alternative or informal financial instruments. Income plays a more significant role. Compared to the poorest quintile, individuals in the top quintiles are substantially less likely to cite cost and distance as barriers. For example, those in the fourth income quintile are 17.8 percentage points less likely to cite distance, and those in the fifth are 40.2 percentage points less likely to cite high costs. Interestingly, higher-income individuals are more likely to cite household saturation (e.g., +22.0 percentage points for the fourth quintile and +18.8 for the fifth), suggesting account sharing may be a rational choice rather than exclusion due to constraints.

Employment status is also a strong predictor. Individuals in the labor force are significantly more likely to mention documentation (+18.3 percentage points) and lack of money (+15.5 percentage points) as barriers, reflecting possible formal requirements and liquidity challenges tied to employment.

In summary, the findings emphasize that barriers to financial inclusion in the Philippines are shaped by a mix of structural (documentation, cost), attitudinal (trust, religion), and demographic (age, gender, income) factors. Crucially, digital financial engagement emerges as a key determinant, reducing trust-related concerns while influencing decisions related to household financial arrangements. These insights suggest that promoting digital financial platform adoption, coupled with policy interventions addressing both voluntary and involuntary exclusion, can help expand access to financial services.

Table 12. Perceived barriers to account ownership and individual characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
VARIABLES	Too far	Too expensive	Lacking docs	Lack trust	Religious reasons	Lack money	Family already has one	No need for financial services
Digital Financial Engagement Index	-0.204 (0.149)	-0.016 (0.140)	0.072 (0.146)	-0.294* (0.155)	-0.006 (0.147)	-0.056 (0.114)	0.355*** (0.123)	0.102 (0.139)
<i>Age group (Youth, 15–24, ref.)</i>								
Young adults (25–34)	0.100 (0.073)	0.104 (0.073)	-0.101 (0.069)	-0.100 (0.065)	0.057 (0.062)	-0.023 (0.062)	-0.049 (0.070)	-0.005 (0.072)
Mid adults (35–44)	0.040 (0.089)	0.174** (0.086)	-0.161** (0.073)	-0.043 (0.080)	-0.012 (0.064)	0.056 (0.064)	-0.158** (0.067)	-0.065 (0.087)
Older adults (45–59)	0.095 (0.082)	0.122 (0.079)	-0.054 (0.080)	-0.071 (0.073)	0.130* (0.073)	-0.031 (0.070)	0.059 (0.075)	-0.030 (0.078)
Senior (60+)	-0.113 (0.119)	-0.096 (0.135)	-0.118 (0.140)	-0.040 (0.108)	0.001 (0.134)	-0.277** (0.128)	0.034 (0.116)	0.019 (0.134)
<i>Gender (Male, ref.)</i>								
Female	-0.061 (0.057)	0.006 (0.058)	-0.051 (0.056)	-0.031 (0.053)	0.086 (0.054)	-0.011 (0.050)	0.131** (0.054)	0.052 (0.058)
<i>Education (Completed primary school or less, ref.)</i>								
Completed secondary school	0.004 (0.062)	-0.021 (0.061)	0.030 (0.059)	-0.055 (0.056)	-0.051 (0.048)	0.044 (0.056)	0.044 (0.055)	-0.010 (0.059)
Completed tertiary education or more	-0.108 (0.099)	-0.090 (0.114)	-0.018 (0.125)	0.145 (0.123)	-0.118 (0.085)	0.078 (0.101)	0.066 (0.099)	0.150 (0.112)
<i>Income quintiles (1st income quintile)</i>								
2 nd income quintile	-0.007 (0.083)	-0.088 (0.083)	0.097 (0.082)	0.051 (0.073)	0.024 (0.069)	0.003 (0.069)	0.049 (0.078)	-0.01 (0.084)
3 rd income quintile	-0.077 (0.086)	-0.094 (0.082)	0.040 (0.078)	0.052 (0.078)	0.057 (0.067)	0.046 (0.076)	-0.041 (0.069)	-0.089 (0.081)
4 th income quintile	- 0.178** (0.085)	-0.108 (0.087)	0.082 (0.087)	0.107 (0.083)	0.100 (0.078)	0.052 (0.076)	0.220*** (0.084)	0.076 (0.089)
5 th income quintile	-0.167* (0.091)	-0.402*** (0.084)	0.032 (0.090)	-0.081 (0.071)	- 0.137*** (0.048)	-0.167* (0.093)	0.188* (0.098)	-0.061 (0.091)
<i>Employment (Out of workforce)</i>								
In the labor force	0.080 (0.061)	0.020 (0.059)	0.183*** (0.055)	0.037 (0.052)	0.007 (0.054)	0.155*** (0.059)	-0.059 (0.052)	0.055 (0.060)
Observations	504	504	504	504	487	504	504	504

Note: Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.1

Source: Authors' estimates

A comparison of the results across ownership, usage, and barriers shows a clear pattern in how digital financial engagement influences financial inclusion. The effect is strongest for account

ownership, where a one-unit increase in Digital Financial Engagement Index is associated with an increase of about 78 percentage points in the likelihood of having an account. The corresponding effects on account usage are positive but smaller, such as 34 percentage points for formal saving and 24 percentage points for formal borrowing. The analysis of barriers helps explain this gap. Individuals may be able to open accounts because digital financial engagement makes access easier, but continued usage depends on structural and behavioral factors such as income constraints, trust in financial institutions, documentation requirements, and household financial arrangements. Overall, digital financial engagement facilitates entry into the financial system, but active and sustained participation is shaped by the broader socioeconomic environment in which individuals operate.

4.2 Supply-Side Analysis

4.2.1 Institutional Trends (IMF Financial Access Survey, 2016-2024)

To assess the evolving supply-side landscape of financial inclusion in the Philippines, Tables 11 and 12 present a comprehensive set of indicators drawn from the IMF Financial Access Survey (FAS), covering the years 2016 to 2024. Table 13 captures the scale and distribution of physical infrastructure, particularly automated teller machines (ATMs) and bank branches, while Table 14 summarizes the volume of credit and deposit services across various types of financial institutions, including commercial banks, cooperatives, other deposit takers (ODTs), and microfinance institutions (MFIs).

As shown in Table 13, physical access to financial services improved incrementally over the period. The number of ATMs nationwide increased from 19,084 in 2016 to a peak of 23,422 in 2022, before declining slightly to 23,352 in 2023 and 23,367 in 2024. In relative terms, ATM density rose from 26.86 to 28.43 per 100,000 adults and from 64.00 to 78.32 per 1,000 square kilometers between 2016 and 2023. This suggests moderate improvements in accessibility, especially in terms of geographic coverage. In contrast, the number of commercial banks remained stable, fluctuating between 42 and 46 institutions. Branch density also saw only marginal growth: from 20.92 to 24.14 per 1,000 square kilometers and from 8.78 to 8.76 per 100,000 adults. Similar patterns were observed among ODTs, such as rural and thrift banks, whose branch densities plateaued after 2020. These figures indicate that while physical infrastructure has expanded slightly, the pace has slowed in recent years.

Table 14 reflects substantial growth in credit and deposit capacities, particularly within the formal banking sector. Between 2016 and 2024, commercial bank loans grew from ₱4.3 trillion to ₱9.6 trillion, while deposits expanded from ₱7.4 trillion to ₱14.8 trillion. This steady increase highlights the resilience and deepening of the formal financial system. Credit unions and ODTs also experienced growth, although at a slower and more uneven pace. Credit union loans rose from ₱5.3 billion to ₱6.8 billion, while deposits increased from ₱9.4 billion to ₱13.4 billion. ODT loans and deposits likewise expanded, reaching ₱840.3 billion and ₱1.12 trillion, respectively, by 2024. Notably, MFIs showed significant momentum. MFI loans grew from ₱0.87 billion in 2016 to ₱6.3 billion in 2024, while deposits nearly tripled from ₱6.7 billion to ₱18.9 billion. These trends point to the growing relevance of MFIs in promoting inclusion among lower-income and underserved populations.

Table 13. Financial Access Infrastructure in the Philippines, 2016–2024

Indicator	2016	2017	2018	2019	2020	2021	2022	2023	2024
ATMs (countrywide , number)	19,084	20,279	21,278	21,780	22,781	22,987	23,422	23,352	23,267
ATMs per 100,000 adults	26.86	27.91	28.65	28.71	29.4	29.1	29.09	28.43	—
ATMs per 1,000 sq. km	64	68.01	71.36	73.05	76.4	77.09	78.55	78.32	—
Number of commercial banks	42	43	45	46	46	46	45	45	44
Commercial bank branches per 1,000 sq. km	20.92	21.74	22.28	23.19	23.57	23.6	24.12	24.14	—
Commercial bank branches per 100,000 adults	8.78	8.92	8.94	9.12	9.07	8.91	8.94	8.76	—
ODT branches per 100,000 adults	5.98	6.21	6.49	6.5	6.37	6.37	6.19	6.75	—
ODT branches per 1,000 sq. km	14.25	15.13	16.15	16.55	16.56	16.86	16.72	18.6	—
Source: IMF Financial Access Survey, 2024; Authors' calculations									

Table 14. Credit and Deposit Services in the Philippines, 2016–2024

Indicator	2016	2017	2018	2019	2020	2021	2022	2023	2024
MFI loans	0.87	1.24	1.68	1.88	2.24	3.63	3.99	4.71	6.26
MFI deposits	6.75	9.20	11.88	13.76	15.36	16.20	16.22	16.88	18.92
ODT loans	673.50	759.93	804.15	759.35	714.94	700.25	597.06	716.33	840.25
ODT deposits	912.06	937.30	965.61	958.57	957.96	1,058.73	837.91	994.54	1,121.77
Credit union loans	5.27	5.71	6.00	6.21	5.78	6.17	7.08	6.93	6.81
Credit union deposits	9.38	10.74	9.32	10.16	10.81	11.78	12.27	12.79	13.37
Commercial bank loans	4,332.65	5,241.35	6,001.55	6,634.62	6,740.36	6,968.92	7,928.37	8,723.53	9,569.77
Commercial bank deposits	7,384.42	8,180.73	8,718.27	9,643.53	10,458.06	11,499.72	12,657.01		

Note: All Values in ₱ Billion**Source:** IMF Financial Access Survey, 2024; Authors' calculations

4.2.2 Key Informant Interviews Insights

4.2.2.1 Profile of Industry and Public Policy Informants

A total of 12 industry and public policy experts were interviewed for this study as summarized in Table 15. The questions revolved around the topics of: (a) State of artificial intelligence and its awareness in the Philippines, (b) State of financial inclusion (c) Profile of financial consumers and their financial behavior and level of financial literacy, and (d) Infrastructure and technology enablers and barriers for AI. The set of questions is crafted based on the findings of quantitative analysis.

Table 15. Profile of Industry and Public Policy Informants

Participant	Expertise/Background	Region
Participant 1 (<i>Adrienne Heinrich, PhD</i>)	Head of AI Center of Excellence UnionBank	National Capital Region
Participant 2 (<i>Ruel Maningas, PhD</i>)	Data Analytics University of Asia and the Pacific (UA&P)	National Capital Region
Participant 3 (<i>Jared Martin Desello, MSIE</i>)	Associate Professor Financial Inclusion and Financial Literacy Japan University of Economics	East Asia
Participant 4	Public Policy	National Capital Region
Participant 5	Universal Bank	National Capital Region
Participant 6	Credit Cooperative	National Capital Region
Participant 7	Savings and Loan Association	Region IV - Calabarzon
Participant 8	Savings and Loan Association	Region IV - Calabarzon
Participant 9	Savings and Loan Association	Region IV - Calabarzon
Participant 10	Savings and Loan Association	Region IV - Calabarzon
Participant 11	Savings and Loan Association	Region IV - Calabarzon
Participant 12	Savings and Loan Association	Region IV - Calabarzon

Source: Authors' compilations

The analysis of the interview transcripts identified key themes related to the above-mentioned topics. Direct quotes were used to support these findings.

4.2.2.2 Current Status of Artificial Intelligence, Data Analytics and Digital Transformation in the Financial Sector

The participants provided their perspectives about AI adoption in the financial sector. One participant commented that *“the financial sector has varying levels of AI adoption among financial institutions, from advanced digital banks to smaller rural banks that are still exploring AI.”* The interviewee shared that this is rooted in the differences in their focus, priorities, and strategies. The interviewee further remarked that *“AI adoption is still very nascent among institutions.”* To offer insights, Dr. Adrienne Heinrich, Head of AI Center of Excellence of UnionBank, mentioned that *“the organization uses a combination of self-created AI solutions and AI solutions provided by partners and vendors.”* Another participant from a Universal Bank stated that the organization also utilizes AI-generated insights from external providers. On the other hand, participants from the Savings and Loan Association expressed that although they are aware of Artificial Intelligence applications, its implementation in the institution has not yet materialized. They cited two reasons: *“their distance from Metro Manila means no one has introduced it to them, and they believe that AI applications are not yet*

necessary for their current market segment.” Furthermore, the participant from the Credit Cooperative expressed interest in leveraging AI and digital tools to improve efficiency and accessibility; however, “...at the moment, *AI is not yet organically needed.*” He highlighted the organization wants to expand services beyond credit to become multi-purpose, with a new grocery initiative aimed at increasing returns and attracting more members.

In terms of data analytics and digital transformation in relation to AI, Dr. Ruel Maningas, the Vice Dean of UA&P School of Management and Program Director of Master in Applied Business Analytics (MABA) Program, stated the importance of digital transformation in local government and microfinance organizations, citing past projects like the Farmers Information Technology Services in the 1990s. He emphasized the need for data governance and quality before implementing analytics and AI, citing examples of microfinance institutions in Cebu and Manila that have automated systems but limited use of analytics. He also touched on the present challenges of managing large amounts of complex data and the importance of establishing a single source of truth for business trust.

The general consensus from participants highlighted that AI applications are utilized for enhancing “Know Your Customer” (KYC and eKYC processes), fraud detection and fighting crimes, transaction monitoring, customer experience, credit scoring, risk management, tracking policy rates and yields, and virtual customer service assistants. One participant commented that “*AI is primarily used for internal processes for work to be more efficient.*” Based on these findings, the institutions appear to be in different stages of AI maturity.

4.2.2.3 Opportunities and Challenges of Artificial Intelligence and Financial Inclusion

The participants also elaborated the potentials and concerns about expanding financial inclusion through AI. Dr. Heinrich shared her experience of moving to the Philippines 6.5 years ago and observing the inefficiencies in financial services compared to Europe. She noted that many Filipinos lack bank accounts and that the pandemic accelerated digital transformation in the financial sector. She also noted that one of the consequences of the pandemic in the Philippines was the high incidence of phishing scams. The National Bureau of Investigation's Cybercrime Division reported in August 2020 a 200% increase in phishing campaigns since the start of the lockdowns in March 2020. This made phishing the top cybercrime committed in the country during that period. She believes that “*this challenge stems from the Philippines being in the nascent stages of its digital transformation and cybersecurity evolution.*” She stated that “*studies suggest greater Digital Financial Engagement leading to both higher digital literacy and awareness of potential scams.*” Moreover, she highlighted that only 2% of Filipinos can correctly answer basic financial literacy questions according to BSP's 2021 Financial Inclusion Survey, emphasizing the need for improved financial and digital education. Lastly, she agreed that financial literacy and digital literacy should be developed together to help Filipinos make informed decisions about managing their money through digital tools. These echo the sentiment that financial literacy, living within means, and become entrepreneurial are key focus areas for education for the Credit Cooperative members.

Mr. Jared Martin Desello of Japan University of Economics who is an expert in financial inclusion and financial literacy compared Japan's efficient banking processes with the Philippines, where cultural barriers and intimidating bank environments can hinder financial inclusion. He observed “*that while targeted financial literacy programs for older adults are common Japan, he had not observed this in the Philippines.*” He also noted that digital platforms have improved access in the Philippines, but the specific needs of customers should

guide the development of financial products and services. Individuals at different life stages would have differing demands on what financial products or services they need. He also mentioned the importance of language settings in financial apps, noting that some customers prefer English while others prefer Filipino. For him, among the characteristics of a good financial inclusion system are ease of use and access. He also stressed the importance of financial literacy and self-responsibility in retirement planning using Japan as an example where government promotes early investment through tax-free incentives and robust digital platforms. Based on his research on retirement savings in the Philippines, he noted that the benefits of the current pension system will be insufficient to cover post-retirement expenses and will rely heavily on individual savings or investment responsibility. He observed that Japan's government has been proactive by setting a recommended savings target of 20 million yen, an amount meant to supplement pension benefits, which encourages individuals plan for retirement. Moreover, he explained that private companies in the Philippines are more active in promoting financial inclusion, often targeting specific demographics like stock investors. He suggested that microfinance institutions and cooperatives could be effective partners for targeting underserved populations.

The participant from a Universal Bank pointed out “...*the digital divide and age-related differences in financial behavior, with ultra-high-net-worth clients still conservative and risk-averse in their investment approaches despite more sophisticated knowledge.*” Furthermore, “*these clients are more likely to transfer between banks for slightly higher yields, emphasizing the importance of branch managers in maintaining client relationships.*” She observed that younger clients are more open to digital financial services. In terms of financial literacy, she explained that while her team focuses on macroeconomic and investment analysis, they have a group within the bank dedicated to provide for financial inclusion, including financial literacy initiatives. She emphasized that financial awareness and proper understanding of products are important even before starting to invest, noting that there is usually a mismatch on the clients' risk appetite and the products that they want to invest in. She also highlighted that gambling is affecting consumer spending patterns and may be affecting household savings. In terms of financial inclusion and digitization challenges in the country, she explained that while many Filipinos use digital wallets like GCash for transactions, traditional banks offer more comprehensive services and investment opportunities, though high maintenance costs and low disposable income remain barriers to broader financial inclusion. There are two points she stressed in relation to these challenges: system automation could lead to job losses in bank operations and the importance of clean, high-quality data for measuring financial inclusion.

The public policy expert reiterated AI's role in financial inclusion and credit scoring, particularly for digital banks. The interviewee acknowledged that “*while AI is a valuable tool, it can potentially exclude vulnerable borrowers due to lack of sufficient data to properly predict the behavior of these types of borrowers.*” The interviewee noted that banks are still learning and improving their credit scoring systems, especially for customers from rural or low-income groups. While AI has potential to enhance financial inclusion, particularly through digital banks targeting unbanked populations, there are still challenges in lending to certain segments and refining credit scoring models for those without traditional financial footprints. Due to these constraints, these people may eventually end up availing services from informal lending institutions. The interviewee pointed out how banks within larger ecosystems are able to gather more comprehensive customer data and refine their credit scoring approaches over time.

The interviewee also emphasized the need for scam prevention and financial literacy education for vulnerable populations. In terms of the challenges on financial inclusion in the Philippines,

particularly why people are hesitant to open formal financial accounts, the interviewee explained that consumer confidence issues, lack of knowledge about digital banking benefits, fraud concerns, and infrastructure limitations, including poor internet connectivity and lack of formal documentation, are key barriers. The interviewee noted that the BSP's digital transformation roadmap for three years (i.e. 2020 to 2023) aims to achieve 70% financial account penetration and 50% digital retail transactions, with initiatives like simplified account-opening processes and e-wallets helping to address these challenges. On the consumer side, the interviewee noted that *“while some consumers are aware of AI through tools like ChatGPT, many others are not aware of AI's presence in financial services, such as automated payment reminders.”* The interviewee discussed the need for greater consumer education about AI and its implications, as well as the development of governance frameworks to guide AI adoption in government institutions. The interviewee emphasized the importance of helping stakeholders understand and adapt to AI technology to avoid displacement in the workforce.

Participants from the Savings and Loans Association acknowledged that AI already exists, and it is a matter of time before they embrace it. However, they still need to upgrade their current infrastructure, and they would want to limit it and still have human intervention as this would affect their employees. They are open to adopting AI, influenced by its use among their private bank partners. However, this adoption is contingent upon two key conditions: improved internet connectivity and tightened cybersecurity measures to ensure their members' peace of mind. The institution digitized its recording, accounting, and membership systems, as well as the conduct of elections, several years ago to comply with BSP requirements. It has an existing IT infrastructure that can support inter-business operations and cater to its well-defined market. However, a significant challenge is internet connectivity. They have observed that the internet is much faster in National Capital Region than in their location. In addition, one of the key officials raised that *“their technology is not as advanced as that of larger financial institutions, especially those in Metro Manila.”*

Drawing from their perspectives, they highlighted major points with regard to artificial intelligence and financial inclusion. They underscore the importance of consumer education concerning AI, digital literacy, and financial literacy - particularly for the vulnerable sector - to ensure these individuals can make informed decisions about managing their money. Furthermore, financial awareness and a proper understanding of products are also crucial before investing, which helps align the client's risk appetite with the products they choose, especially for those in the banked segment. The development of financial products and services should be guided by the specific needs of customers across different life stages and socio-economic backgrounds. Moreover, human relationships remain necessary to maintain client connections, and there is a need for development of governance frameworks to guide AI adoption in government institutions. On the other hand, one pointed out that there should be clean, high-quality data for measuring financial inclusion. Another participant noted that people exposed to AI tend to have higher digital literacy and greater awareness of potential scams.

Regarding challenges, they recognized that cybercrimes, the digital divide, age-related differences, conservatism and risk-averse behaviors deter individuals from taking advantage of banking services. The key barriers leading to consumer confidence issues are lack of knowledge about digital banking benefits, lack of funds, fraud concerns, and infrastructure limitations. There is also a risk of data and algorithms being biased against unbanked populations in digital banking.

4.2.2.4 Current Initiatives on Artificial Intelligence and Financial Inclusion

The participants presented their institutions projects to promote financial inclusion, and leveraging AI. Dr. Heinrich noted that while many households interact with AI daily through various services, UnionBank makes a deliberate effort to ensure their AI-powered features are visible and accessible to customers. This includes initiatives showcased on their website and the implementation of their Explainable & Responsible AI Toolkit. She highlighted the bank's proactive approach in reducing discrimination in lending decisions by leveraging AI. She also emphasized UnionBank's commitment to engaging customers directly with AI-driven services while maintaining a high level of transparency regarding how AI is utilized.

Dr. Heinrich outlined four key components of readiness for digital financial services: digital access, financial literacy and trust, active usage, and an enabling environment. She explained that UnionBank is addressing these with multiple initiatives, one of these being the *Bank on Wheels*, which provides mobile banking services to rural areas and disaster-affected regions. The bank is also actively enhancing customer support and lowering entry barriers for new users.

She also emphasized the importance of a human-centric approach in designing AI features and products. She highlighted UnionBank's commitment to responsible AI through an explainable and responsible AI toolkit that assesses AI models for bias and transparency before deployment. The bank also prioritizes building customer trust by being transparent about their AI practices and actively engaging with users to address their concerns. Additionally, she discussed exploring alternative data sources to expand access to financial services for marginalized groups in the Philippines.

Also, UnionBank is dedicated to enhancing financial inclusion in the Philippines, including rural and low-income populations. Dr. Heinrich mentioned how UnionBank, through its fintech subsidiary UBX, leverages the blockchain-powered i2i platform to deliver essential cash and financial services to remote and underserved areas across the Philippines. This initiative has significantly advanced financial inclusion by connecting rural banks and community-based financial institutions to the broader financial ecosystem. Additionally, the bank has implemented AI in customer service to enhance efficiency. She also touched on the importance of word-of-mouth referrals in building customer trust and the bank's strategy to cater to different segments through subsidiaries like City Savings Bank, which targets specific groups such as public-school teachers and government workers. UnionBank's subsidiaries include Banko Kabayan, which serves rural and semi-urban areas, PetNet for remittances targeting OFWs and small businesses, and UnionDigital, a digital bank focused on underserved segments without minimum balance requirements.

According to the participant from the other universal bank, the institution aims to increase its banking services to reach 100% of Filipinos, particularly concentrating on underserved populations. This is possible through their rural bank subsidiaries which manages various branches and kiosks in provinces or remote areas. The participant stated that *"their extensive network of branches, which is largest among its competitors, and emphasized the importance of physical branches in building trust among Filipinos, especially in remote areas."*

Based on the above-mentioned initiatives, universal banks are more equipped and aggressive in incorporating artificial intelligence in their services and promoting financial inclusion especially in remote and underserved areas.

4.2.2.5 The Way Forward: Harnessing Artificial Intelligence for Financial Inclusion

Dr. Maningas emphasized the importance of data quality and analytics in decision-making, emphasizing that actionable insights derived from high-quality data can support effective interventions. He highlighted the need to validate analytics results and the role of AI in predictive analytics for identifying at-risk groups. He suggested that one of the most effective interventions after identifying such groups is considering factors like cost of production and competitor pricing. He also highlighted the drivers of AI and data analytics in financial inclusion, including technology availability, stakeholder recognition, industry guidance, and policy development. He emphasized the importance of data quality and ethics in AI implementation, urging government and private sector support for training and capacity building. Finally, he shared insights on how data analytics and AI can support, but not replace, human decision-making in the financial sector. He emphasized the importance of digital transformation and digitization in companies and institutions, highlighting the use of AI tools for data processing and analysis. He also mentioned advocating for digital transformation at the local government level to improve data management and decision-making.

Dr. Heinrich explained that trust in AI requires providing theoretical knowledge about its capabilities and potential risks, such as deepfakes and biased decision-making. She emphasized the importance of practical experience with AI and guidance from tutors or communities to build trust. She also highlighted the gender gap in accessing loans in the Philippines and how AI can contribute to fairer treatment if properly trained. Driven by the mission to broaden financial access for all Filipinos, UnionBank is proactively seeking innovative approaches to foster greater financial inclusion across the country. She emphasized the need for public-private partnerships, improved internet and electricity access in remote areas, and educational programs to help underserved areas and population groups like older generations embrace digital banking. Dr. Heinrich suggested that the government could play a role in developing three-phase programs combining theoretical learning, hands-on experience, and community guidance to promote digital financial inclusion.

Participants from the Savings and Loans Association wish to cater to reach more clients to avail of their services. However, they expressed concerns that fulfilling numerous requirements and adhering to various guidelines and policies serves as a barrier to business expansion and increasing their membership nationwide. They hope that *“the government will be more flexible in this regard”*. Additionally, they fear growing competition from large players and other Savings and Loan Institutions within their market segment. They are also concerned about the potential influence of larger financial competitors in shaping these guidelines and policies, which they worry may be biased in favor of these major rivals. The participant from the Credit Cooperative also pointed out the tedious government requirements.

For Mr. Desello, he noted that he had not seen significant use cases of AI in educational platforms for financial literacy, with most training still relying on human instructors. With all these observations, he aspires that *“AI can be used as leverage to broaden the financial education and target the underserved population in the Philippines.”* For the participant from the Universal Bank, she stressed the challenges and importance of implementing industry roadmaps particularly for government initiatives and financial inclusion projects. She emphasized *“the need for stricter execution and widespread awareness to ensure effective outcomes.”*

Finally, according to the public policy expert, there is a need for a whole-of-government approach to address AI awareness across sectors. He highlighted the importance of coordinating with other government agencies to develop an AI education roadmap, similar to existing cybersecurity initiatives.

In summary, the participants shared salient points in leveraging AI toward financial inclusion. First, quality of data, technology availability, stakeholder recognition, industry guidance, and policy development are the primary drivers of AI and data analytics for enhancing financial inclusion. Second, providing practical experiences with AI and guidance from tutors can increase trust in AI. Third, there is a need for public-private partnerships to improve internet and electricity access in remote areas and develop education programs that help underserved areas and older population; this partnership should also focus on developing programs for enhancing digital financial inclusion and knowledge about AI. Fourth, relaxing government requirements is necessary to facilitate expanding the reach of financial institutions. Fifth, AI should also be used for financial education. Lastly, there should be the development of an AI education roadmap and stronger enforcement of that roadmap.

5. Discussion and Conclusion

This study addresses a critical research gap in the Philippine literature by examining how digital financial engagement influences financial inclusion from both the demand- and supply-side perspectives. Earlier studies have primarily concentrated on demographic and socio-economic determinants of financial inclusion (Llanto 2015; Llanto and Rosellon 2017; Debuque-Gonzales and Corpus 2021; Debuque-Gonzales and Corpus 2024) or on measuring financial inclusion indices (Debuque-Gonzales and Corpus 2021; Alinsunurin and Alinsunurin 2015). Meanwhile, research on AI has largely focused on the supply-side applications of technology in the financial sector, highlighting the roles of chatbots, credit scoring, and risk management systems (Kaur and Dharmadhikari 2023; Mhlanga 2020; Kandpal and Khalaf 2020). Few empirical studies have examined how AI awareness, digital readiness, and consumer behavior interact to shape financial inclusion outcomes in the Philippines (Carandang 2023; Lim and Parrocho 2024; Jaca et al. 2025; Bragas et al. 2025). This study responds to this gap by examining how digital financial platform engagement influences financial inclusion and how AI is being deployed within these platforms to enhance accessibility and security in digital transactions. It combines nationally representative data from the World Bank Global Findex (2021), institutional indicators from the IMF Financial Access Survey (2016–2024), and qualitative insights from financial institutions and policy experts, thereby providing empirical evidence of how digital financial engagement shapes financial inclusion outcomes, while documenting AI deployment patterns within the financial sector.

5.1 Demand Side Evidence on Digital Financial Engagement

Results from the demand-side analysis reveal that digital financial engagement is a strong and consistent determinant of financial inclusion. The constructed Digital Financial Engagement Index shows that individuals who actively use digital financial platforms are significantly more likely to own and use formal financial accounts. While digital financial engagement is positively associated with a wide range of financial activities, the magnitude of the effect becomes more modest for more complex behaviors such as saving, borrowing, and digital transactions. These findings are consistent with international evidence (Fazal et al. 2023;

Noreen et al. 2023; Lim et al. 2025), which emphasizes that engagement with digital financial platforms is a critical enabler of financial participation.

However, several barriers persist. The lack of money, high perceived costs, documentation issues, and low trust remain key constraints among Filipino adults, particularly among low-income groups. Interestingly, the regression results among unbanked respondents indicate that higher digital financial engagement significantly reduces the likelihood of citing “lack of trust” as a reason for not owning an account. This suggests that greater familiarity with digital financial platforms may help build confidence in formal financial institutions and digital transactions. The results also reveal that while the gender gap in account ownership is narrowing, there remain persistent socioeconomic and generational divides, where younger, more educated, and higher-income groups exhibited greater financial engagement compared to older or poorer populations. Despite the Philippines’ broad access to mobile and internet networks, digital financial engagement remains uneven, suggesting that connectivity alone does not guarantee inclusion. This gap emphasizes the importance of translating digital access into effective and trusted financial usage.

Taken together, these results show that digital financial engagement does not affect all stages of financial inclusion in the same way. It seems to matter most at the point of entry by helping people open accounts and get connected to digital financial services. But once people already have an account, its influence becomes weaker. More complex behaviors such as saving regularly, borrowing, or using accounts for transactions still depend heavily on a person’s income situation, confidence in the system, and ability to meet requirements. Thus, while digital financial engagement opens the door to participation, the extent of sustained usage remains shaped by the socioeconomic environment in which individuals operate.

5.2 Supply Side Evidence on Financial Infrastructure and AI Deployment

From the supply side, institutional data from the IMF Financial Access Survey (2016–2024) and insights from KIIs reveal that financial infrastructure has expanded moderately, with growth in the number of ATMs, deposits, and loans, particularly within commercial banks and microfinance institutions. Yet, AI adoption remains concentrated among large, digitally advanced financial institutions, while smaller cooperative and savings and loan associations continue to face significant barriers.

KII participants consistently observed that AI adoption in the Philippine financial sector is still at a nascent stage, with institutions exhibiting varying levels of digital maturity. Universal banks have made significant progress in applying AI for customer engagement, risk management, and fraud detection, while rural and community-based financial entities continue to face constraints stemming from poor internet connectivity, cybersecurity risks, and regulatory burdens. As emphasized by Dr. Adrienne Heinrich (UnionBank), the key challenges lie in ensuring data quality, algorithmic fairness, and public trust in AI systems. Other key informants reiterated that the digital divide, infrastructural gaps, and lack of digital literacy continue to hinder smaller institutions from fully integrating AI into their operations, even as they recognize its potential to improve accessibility and operational efficiency.

Amid these challenges, both industry and policy experts acknowledged that AI holds tremendous potential to enhance inclusion through improved credit scoring, fraud prevention, and operational efficiency, provided that strong governance frameworks and responsible AI practices are in place. These insights reinforce the literature’s argument that AI can bridge

information asymmetries and extend financial services to the unbanked (Telukdarie and Mungar 2023; Bazarbash 2019), but only when institutions and consumers are equally equipped to engage with technology.

5.3 Convergence of Demand Side and Supply Side Evidence

The convergence of demand- and supply-side evidence highlights that digital financial engagement serves both as a driver and an indicator of financial inclusion. On the demand side, greater digital engagement empowers consumers to use financial services more confidently and effectively. On the supply side, the readiness of financial institutions such as banks and cooperatives, reflected in their technological capabilities, AI deployment strategies, governance frameworks, and risk management systems, determines how accessible, affordable, and secure these services become. These findings reaffirm the literature's call for a balanced, human-centered approach to digital financial transformation that combines technological innovation with education, trust-building, and inclusive system design.

Importantly, while our demand-side analysis measures digital financial platform engagement rather than direct AI exposure or interaction, the connection to AI is established through our supply-side findings. The digital behaviors captured in our index, including mobile payments, online banking, digital transactions, align precisely with the platforms where financial institutions are actively deploying AI technologies. Universal banks and major e-wallet providers reported using AI for fraud detection in mobile transactions, credit scoring for online loan applications, chatbots for digital customer service, and transaction monitoring in online banking. As a results, individuals with higher scores on our Digital Financial Engagement Index are more likely to be interacting with AI-enhanced services, even if they do not recognize these features as AI-powered. Thus, digital financial engagement serves as both the behavioral foundation for financial inclusion and the channel through which individuals encounter AI-enhanced financial services.

5.4 Policy Alignment and Implications

The study also highlights the need for stronger alignment between the National Strategy for Financial Inclusion (NSFI 2022–2028) and the National AI Strategy Roadmap 2.0. While the NSFI focuses on access and affordability, the AI roadmap emphasizes innovation and competitiveness. Integrating these frameworks would help ensure that efforts to promote AI-driven financial inclusion go hand in hand with inclusive AI governance, preventing technology from widening existing social and economic inequalities.

Overall, this study provides new empirical evidence that digital financial engagement significantly enhances financial inclusion in the Philippines, demonstrating that digital literacy and active platform usage have become prerequisites for equitable participation in the financial system. However, technology alone is not enough to achieve inclusion. Persistent socioeconomic barriers, institutional disparities, and gaps in digital literacy continue to limit participation, calling for stronger collaboration among policymakers, regulators, and the private sector. A more inclusive financial landscape in the age of AI should be human-centric, data-driven, and equity-oriented, anchored on digital literacy, strong infrastructure, and responsible innovation. Strengthening public–private partnerships, improving data governance, and ensuring transparency in AI adoption are critical to building consumer trust and fostering meaningful participation in the financial system. By closing the gap between

access and trust, the Philippines can transform digital financial platforms, enhanced by AI technologies, into a true catalyst for inclusive and sustainable growth.

6. Recommendations

Given these findings, several actionable measures are proposed to strengthen financial inclusion through enhanced digital financial platforms and responsible AI deployment in the Philippines:

Government The evidence shows that digital financial platform engagement is a critical determinant of financial inclusion, and that AI technologies can enhance the safety, efficiency, and accessibility of these platforms. To maximize its potential, the government must prioritize enhancement of internet connectivity and electricity access, particularly in rural and geographically isolated areas. Reliable digital infrastructure is the foundation upon which both consumer engagement with digital platforms and institutional AI deployment depend. Ensuring digital networks are fast, reliable, and secure will enable more Filipinos to access digital financial services and benefit from AI-driven financial services.

Public-private partnerships can play a vital role in expanding digital infrastructure and supporting AI-related initiatives. At the same time, laws on data protection, cybersecurity, and digital trust – including the Data Privacy Act and Cybercrime Prevention Act- should be regularly reviewed and strengthened to ensure stronger and more effective implementation and enforcement.

Since trust remains a cornerstone of digital adoption, the government should invest in initiatives that enhance public confidence in AI and financial technologies. Developing national roadmaps for AI education and capacity-building would also be beneficial, as increased awareness and literacy can empower citizens to use these tools responsibly.

Moreover, AI-related indicators, such as AI literacy, usage levels, and public trust, can be integrated into national surveys to generate reliable data for policymaking. These datasets can be used in future studies on the impact of AI on the growth and inclusiveness of the financial sector. The study recommends integrating AI-related questions in future rounds of BSP's Financial Inclusion Surveys to measure directly consumers' AI awareness, knowledge and usage. Specific questions could include: *"Are you aware that the digital financial services you use may incorporate AI technologies?"*; *"Do you trust AI systems to help detect fraud in your transactions?"*; and *"Which specific digital platforms do you use most frequently for financial transactions?"* Such data would enable future research to differentiate between high-AI and low-AI platforms and provide more granular insights into AI's role in financial inclusion.

Education AI adoption in the Philippines remains at an early stage and many consumers use AI-powered tools daily, such as chatbots, credit scoring systems, and digital assistants without realizing it. Other individuals are not even aware that these tools exist. This "invisible AI" paradox means that individuals may benefit from or be affected by AI without understanding how it works or how to use it effectively. This lack of awareness underscores the need for education programs that build both financial and AI literacy, particularly among vulnerable groups (i.e. older population and underserved areas)

The government, educational institutions, and private sector should collaborate to design community-based training programs that provide practical exposure to AI applications in

financial services. There can also be programs that focus on underserved populations and older demographics, who often face greater barriers to digital inclusion.

Furthermore, integrating financial and AI literacy into school curricula and workplace training can prepare Filipinos to adapt to the evolving financial landscape. Partnerships with banks and fintech companies can help deliver hands-on workshops and public information campaigns that clarify AI, promote responsible use, and build consumer trust. Programs that teach individuals how to identify fraudulent schemes, safeguard personal data, and understand algorithmic decision-making can empower citizens to engage responsibly with AI-driven financial services.

Industry and Private Financial Institutions Financial institutions in the Philippines have begun incorporating AI into their operations, ranging from customer service chatbots, and fraud detection systems to predictive credit scoring and personalized recommendations. However, our research reveals significant heterogeneity in adoption: while universal banks and major e-wallet providers have achieved advanced AI deployment, smaller rural banks, cooperatives, and savings/loan associations face substantial barriers related to infrastructure, expertise, and resources. In addition, public scepticism and concerns about data privacy, algorithmic bias, fraud, and cybersecurity continue to hinder widespread adoption.

Financial institutions should prioritize responsible AI deployment that addresses these concerns. This includes: (1) ensuring algorithmic transparency and explainability, particularly in credit scoring and lending decisions that affect vulnerable populations; (2) implementing robust testing for bias and discrimination in AI models; (3) providing clear communication to customers about when and how AI is being used in their interactions; and (4) maintaining human oversight for high-stakes decisions. UnionBank's Explainable and Responsible AI Toolkit, highlighted by our key informant, provides a valuable model for other institutions to follow.

To build consumer trust, banks and fintech companies must strengthen security frameworks and enhance transparency. Clear communication about data usage, coupled with strong consumer protection protocols, can help build long-term trust in digital financial platforms and AI-enhanced services.

In parallel, the government's support in upgrading digital infrastructure and enforcing AI-related regulations is essential to sustain these initiatives. Collaborative efforts among regulators, technology providers, and financial institutions will ensure that AI adoption is responsible, inclusive, and aligned with national development goals.

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Appendices

Appendix A. Variables Used in Financial Inclusion Studies

Author(s) (Year)	Title	Data Set	Variables Used	General Findings
Anyangwe et al. (2022)	Determinants of Financial Inclusion: Does Culture Matter?	World Bank Global Index 2017; 85 countries (developing and developed)	Account ownership, saving, borrowing. Hofstede's six cultural dimensions; Controls: GDP per capita, financial sector indicators.	Cultural traits influence inclusion. High power distance and uncertainty avoidance reduce it; individualism and long-term orientation enhance it.
Dar and Ahmed (2021)	Financial Inclusion Determinants and Impediments in India: Insights from the Global Financial Inclusion Index	Global Index (2017) – India sample	Financial inclusion index. Education, income, gender, urban/rural, employment, infrastructure.	Education, income, and urban residence increase inclusion; rurality and low education hinder it.
Debuque-Gonzales and Corpus (2021)	Understanding and Measuring Financial Inclusion in the Philippines	Demand- and supply-side Philippine data (Findex + subnational indicators)	Account ownership, access. Education, income, urbanization, outreach (branches/ATMs).	Inclusion improves with education, income, and access; poorer, less educated households remain excluded.
Debuque-Gonzales and Corpus (2024)	Understanding Financial Inclusion in the Philippines	Philippine Financial Inclusion Survey (2021) and demand-supply indicators	Financial inclusion index (account ownership, usage). Education, income, geography, gender, digital access.	Education, income, and digital access improve inclusion; regional and gender disparities remain significant.
Demirgüç-Kunt et al. (2022)	The Global Findex Database 2021	Global Index 2021 survey – 128,000 adults in 121+ economies	Account ownership, digital payments, saving, borrowing. Demographics, digital access, employment.	Global account ownership reached 76%; digital payments surged during COVID-19; gender and income gaps persist.
Desello (2024)	Constructing a Multidimensional Index for Financial Well-Being	Global Index 2021; cross-country PCA analysis	Financial well-being index. Financial confidence, resilience, macro factors.	Proposes a composite index; negatively correlated with poverty and inequality.
Desello and Agner (2023)	Financial Inclusion and the Role of Financial Literacy in the Philippines	2019 Financial Inclusion Survey (Philippines)	Account ownership, formal service use. Financial literacy, education, income.	Financial literacy, education, and income enhance inclusion among Filipinos.
Jalloh (2023)	Determinants of Financial Inclusion in Sierra Leone	Global Index (2017–2021) – Sierra Leone	Account ownership, formal service use. Education, income, gender, urban/rural, mobile access.	Education, income, and urban residence increase inclusion; gender and rural gaps persist.

Nandru et al. (2021)	Determinants of Digital Financial Inclusion in India	Global Findex (2017) – India subsample	Digital finance use. Mobile ownership, education, income, urbanity, age, gender.	Digital inclusion rises with mobile ownership, education, and income; demographic divides persist.
Ozili (2021)	Financial Inclusion: Globally Important Determinants	Cross-country secondary data and literature review	Financial inclusion indicators. Financial infrastructure, tech access, literacy, policy, culture.	Infrastructure, digital access, and literacy are key global drivers of inclusion.
Quimba et. al (2024)	Readiness for AI Adoption of Philippine Business and Industry: The Government's Role in Fostering Innovation- and AI-Driven Industrial Development	Philippine establishment-level data (PSA SICT, PIDS surveys)	AI readiness index AI watch index AI preparedness index Asia-Pacific AI readiness Index	AI adoption (~15%) concentrated in urban areas and larger firms; barriers include skills gaps and funding; policy recommendations focusing on infrastructure development, human capital development, regulatory frameworks, public-private partnerships, and ethical guidelines.
Razzaq et al. (2024)	Determinants of Financial Inclusion Gaps in Pakistan and Implications for Achieving SDGs	Global Findex 2021 – Pakistan sample	Dep.: Account ownership, service usage (by gender). Ind.: Gender, education, income, employment, digital access.	Large gender gap; education, income, and digital access reduce disparities.

Source: Authors' compilations

Appendix B. Detailed Results of the Multiple Correspondence Analysis (MCA)

Table B1. Eigenvalues and Explained Inertia

Dimension	Principal Inertia	Percent of Inertia	Cumulative Percent
1	0.1354	93.92	93.92
2	0.0002	0.17	94.09
Total	0.1441	100.00	—

Source: Authors' estimates

Table B2. Category Coordinates, Squared Correlations, and Contributions

Variable (Category)	Mass	Coord (Dim 1)	Sqcorr (Dim 1)	Contrib (Dim 1, %)	Coord (Dim 2)	Sqcorr (Dim 2)	Contrib (Dim 2, %)
mobileowner = yes	0.120	0.057	0.871	0.00	0.401	0.079	0.02
mobileowner = no	0.005	-1.472	0.871	1.00	-10.447	0.079	50.5
internetaccess = yes	0.109	0.199	0.977	0.40	0.659	0.019	4.7
internetaccess = no	0.016	-1.381	0.977	3.0	-4.570	0.019	32.9
anydigpayment = no	0.057	-1.152	0.978	7.5	-0.087	0.000	0.0
anydigpayment = yes	0.068	0.954	0.978	6.2	0.072	0.000	0.0
fin14_1 = yes	0.022	2.357	0.929	12.3	-1.312	0.001	3.8
fin14_1 = no	0.103	-0.507	0.929	2.6	0.282	0.001	0.8
fin14a = yes	0.038	1.847	0.923	12.9	-0.511	0.000	1.0
fin14a = no	0.087	-0.799	0.923	5.6	0.221	0.000	0.4
fin14a1 = yes	0.044	1.481	0.963	9.6	-0.172	0.000	0.1
fin14a1 = no	0.081	-0.801	0.963	5.2	0.093	0.000	0.1
fin14b = yes	0.058	1.134	0.974	7.5	0.242	0.000	0.3
fin14b = no	0.067	-0.989	0.974	6.5	-0.211	0.000	0.3
merchantpay_dig = no	0.092	-0.747	0.895	5.1	0.292	0.000	0.8
merchantpay_dig = yes	0.033	2.094	0.895	14.4	-0.818	0.000	2.2

Source: Authors' estimates

Appendix C. Digital Financial Engagement Index and its Key Indicators by Socio-Economic Group

Socio-Economic Group	Owns Mobile	Has Internet	Digital Payment	Used mobile for Finance	Paid bills online	Sent money online	Merchant Payment	AI Awareness Index
Sex								
Female	0.938	0.767	0.393	0.101	0.177	0.240	0.137	0.325
Male	0.903	0.776	0.479	0.165	0.236	0.303	0.236	0.386
Age Group								
15–24 Youth	0.904	0.840	0.340	0.115	0.193	0.258	0.139	0.339
25–34 Young Adults	0.948	0.847	0.574	0.187	0.284	0.371	0.251	0.437
35–44 Mid Adults	0.920	0.791	0.499	0.160	0.233	0.310	0.235	0.391
45–59 Older Adults	0.935	0.660	0.403	0.092	0.184	0.181	0.161	0.301
60+ Seniors	0.868	0.540	0.268	0.064	0.005	0.143	0.083	0.207
Education Level								
Primary or Less	0.849	0.604	0.255	0.056	0.103	0.178	0.074	0.236
Secondary	0.954	0.829	0.435	0.105	0.157	0.247	0.145	0.344
Tertiary or More	0.974	0.940	0.778	0.350	0.531	0.508	0.500	0.609
Income Quintile								
Poorest 20%	0.863	0.672	0.294	0.054	0.096	0.115	0.039	0.236
Second 20%	0.898	0.622	0.263	0.069	0.110	0.180	0.090	0.246
Middle 20%	0.913	0.795	0.332	0.053	0.140	0.240	0.092	0.303
Fourth 20%	0.954	0.812	0.510	0.123	0.195	0.284	0.142	0.368
Richest 20%	0.977	0.953	0.771	0.362	0.486	0.532	0.559	0.616
Employment Status								
Not in Workforce	0.903	0.761	0.312	0.089	0.144	0.226	0.121	0.297
In Workforce	0.930	0.776	0.491	0.152	0.234	0.291	0.214	0.381
Overall	0.921	0.771	0.435	0.132	0.206	0.270	0.185	0.354

Source: Authors' calculations

Appendix D. Formal Financial Account Ownership by Socio-Economic Group in the Philippines

Mean proportion of adults reporting account ownership, debit/credit card ownership, and mobile money use

Socio-Economic Group	Formal Account Ownership	Debit Card Ownership	Credit Card Ownership	Mobile Money Use
Sex				
Female	0.409	0.241	0.180	0.189
Male	0.514	0.359	0.230	0.248
Age Group				
15–24 (Youth)	0.343	0.178	0.123	0.169
25–34 (Young Adults)	0.561	0.361	0.264	0.355
35–44 (Mid Adults)	0.553	0.399	0.193	0.249
45–59 (Older Adults)	0.419	0.282	0.232	0.159
60+ (Seniors)	0.446	0.328	0.214	0.040
Education				
Primary or Less	0.286	0.139	0.108	0.120
Secondary	0.452	0.261	0.153	0.197
Tertiary or More	0.811	0.699	0.329	0.458
Income Quintile				
Poorest 20%	0.297	0.161	0.043	0.093
Second 20%	0.309	0.145	0.081	0.131
Middle 20%	0.375	0.207	0.147	0.144
Fourth 20%	0.543	0.311	0.134	0.225
Richest 20%	0.772	0.666	0.362	0.492
Employment Status				
Not in Workforce	0.350	0.202	0.123	0.168
In Workforce	0.510	0.343	0.233	0.240
Overall	0.460	0.298	0.207	0.217

Note: Values represent survey-weighted means for binary outcomes (1 = yes, 0 = no). Indicators reflect access to and use of financial services.

Source(s): World Bank Global Findex Database 2021; Authors' calculations using weighted survey data.

Appendix E. Formal Financial Account Usage by Socio-Economic Group in the Philippines

Mean proportion of adults reporting usage of formal financial services

Socio-Economic Group	Saving	Borrowing	Remittances	Mobile Transactions	Online Transactions
Sex					
Female	0.147	0.121	0.184	0.256	0.421
Male	0.239	0.144	0.237	0.317	0.491
Age Group					
15–24 (Youth)	0.138	0.085	0.153	0.242	0.495
25–34 (Young Adults)	0.258	0.141	0.250	0.398	0.559
35–44 (Mid Adults)	0.248	0.183	0.265	0.318	0.489
45–59 (Older Adults)	0.166	0.170	0.227	0.229	0.343
60+ (Seniors)	0.108	0.058	0.113	0.161	0.204
Education					
Primary or Less	0.071	0.118	0.119	0.144	0.290
Secondary	0.148	0.116	0.193	0.263	0.463
Tertiary or More	0.538	0.203	0.426	0.615	0.750
Income Quintile					
Poorest 20%	0.040	0.105	0.119	0.124	0.282
Second 20%	0.090	0.128	0.141	0.144	0.312
Middle 20%	0.143	0.146	0.135	0.204	0.475
Fourth 20%	0.125	0.123	0.200	0.325	0.483
Richest 20%	0.558	0.158	0.452	0.630	0.719
Employment Status					
Not in Workforce	0.104	0.082	0.171	0.210	0.385
In Workforce	0.232	0.155	0.227	0.320	0.487
Overall	0.192	0.132	0.210	0.286	0.455

Note: Values represent the mean proportion of adults who reported using formal financial services.

Source(s): World Bank Global Findex Database 2021; Authors' calculations using weighted survey data

Appendix F. Barriers to Account Ownership by Socio-Economic Group

Socio-Economic Group	Too Far	Too Expensive	Lack of Documents	Lack of Trust	Religious Reasons	Lack of Money	Family Has Account	No Need for Services
Sex								
Female	0.445	0.572	0.367	0.239	0.137	0.755	0.270	0.331
Male	0.391	0.600	0.348	0.224	0.197	0.756	0.421	0.400
Age Group								
15–24 (Youth)	0.374	0.496	0.398	0.260	0.124	0.749	0.340	0.353
25–34 (Young Adults)	0.462	0.606	0.361	0.168	0.200	0.782	0.337	0.395
35–44 (Mid Adults)	0.437	0.697	0.270	0.236	0.127	0.850	0.173	0.320
45–59 (Older Adults)	0.494	0.623	0.388	0.235	0.266	0.767	0.374	0.352
60+ (Seniors)	0.269	0.566	0.282	0.282	0.000	0.488	0.532	0.413
Education								
Primary or Less	0.446	0.641	0.341	0.270	0.200	0.735	0.302	0.362
Secondary Education	0.421	0.552	0.379	0.184	0.140	0.770	0.349	0.340
Tertiary or More	0.259	0.448	0.327	0.332	0.089	0.781	0.470	0.520
Income Quintile								
Poorest 20%	0.500	0.718	0.299	0.207	0.129	0.742	0.268	0.354
Second 20%	0.485	0.589	0.389	0.249	0.183	0.738	0.306	0.358
Middle 20%	0.419	0.572	0.362	0.246	0.176	0.816	0.213	0.282
Fourth 20%	0.301	0.553	0.399	0.282	0.231	0.792	0.517	0.474
Richest 20%	0.279	0.265	0.347	0.099	0.008	0.599	0.542	0.341
Employment Status								
Not in Workforce	0.371	0.567	0.261	0.224	0.126	0.650	0.375	0.321
In Workforce	0.451	0.595	0.417	0.237	0.185	0.818	0.313	0.385
Overall	0.421	0.585	0.359	0.232	0.163	0.755	0.336	0.361

Note: This appendix presents weighted averages of reported barriers to account ownership. Barriers include distance, cost, documentation, trust, religion, lack of money, redundant accounts, and perceived lack of need. Values represent proportions and were computed using survey weights.

Source(s): World Bank Global Findex Database 2021; Authors' calculations using weighted survey data.

Appendix G. Key Informant Interview Questions

1. How is the state of Artificial Intelligence (AI) usage in the financial sector? How about in your current institution?
2. Based on your experience, what do you see as the most persistent drivers and barriers to financial inclusion in the Philippines today?
3. How has the growth of digital financial services changed access for marginalized groups—particularly older adults, women, or low-income populations?
4. Are financial institutions equipped with the necessary infrastructure and technology to facilitate digital financial services and capture the benefits of AI?
5. Given the strong link found between AI exposure and both account ownership and usage, what are your institution's current efforts in promoting digital and financial literacy, and how might we align these with financial inclusion objectives?
6. Our findings highlight a persistent digital divide by age. What targeted measures could be implemented to improve fintech usage among older adults?
7. Given that AI exposure appears to reduce mistrust in financial institutions, what initiatives has your institution/agency considered to improve trust and digital confidence among the banked and unbanked?
8. The results suggest that higher income and tertiary education are associated with more formal account usage, especially for savings and credit. How can financial products be restructured to better serve low-income, less-educated groups?
9. How do you interpret the finding that many users defer individual account ownership because someone else in the household already has one?
10. How are you currently addressing barriers related to documentation, trust, and perceived cost, especially for lower-income clients?
11. How transparent is your institution with its customers/clients regarding the use of AI?
12. What are the key gaps in digital and AI-related literacy data that hinder better targeting of financial inclusion interventions?
13. How can predictive analytics or AI tools help identify populations at risk of digital exclusion before they become financially excluded?
14. How do you recommend measuring “digital readiness” for financial services at the household or community level?
15. What policy, product, or partnership would you most like to see implemented in the coming years to accelerate inclusive digital finance and the use of AI?

Appendix H. Key Informant Interview Consent Form

Digital Financial Platform Engagement and Financial Inclusion in the Philippines: Insights on AI Deployment and Policy Implications

Main Research Question: *How does engagement with digital financial platforms influence financial inclusion in the Philippines, and how can artificial intelligence be strategically deployed within these platforms to enhance accessibility, safety, and security?*

Statement of Consent

In the final research paper, I authorize the Research Team to (please check):

- ☐ Attribute me by my full name, designation, and affiliation
Full name: _____
Designation: _____
Company/Affiliation: _____
- ☐ Attribute me with complete anonymity

I am also aware that prior to the publication's submission, the Research Team will provide me with a draft of my statement, allowing me to review the proposed text and verify that my commentary/remarks/insights have been accurately captured.

Name and Signature of Participant

Date